Numerical Integration of Stochastic Differential Equations



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These lecture notes are based on the *Numerical Integration of Stochastic Differential Equations* course given at EPFL for the academic year 2024/2025 by Prof. Fabio Nobile.

Chapter 1

Prerequisites

In this section, we aim to present the main prerequisites for the course, where we review concepts and definitions in Probability Theory along with some basics and more advanced ideas in Stochastic Processes and Stochastic Calculus.

1.1 Probability Theory

This first section is a review on some concepts in Probability Theory, and will mainly consist of a list of definitions and properties.

Definition 1.1.1. *A **probability space** is a triplet $(\Omega, \mathcal{F}, \mathbb{P})$ where:*
1) Ω is the set of outcomes, $\omega \in \Omega$.
2) $\mathcal{F}$ is the σ-algebra: collection of subsets of Ω such that (i) $\varnothing, \Omega \subset \mathcal{F}$, (ii) $A \in \mathcal{F} \Longrightarrow A^c \in \mathcal{F}$, (iii) $\{A_i\}_{i\in\mathbb{N}} \subset \mathcal{F} \Longrightarrow \bigcup_{i=1}^{\infty} A_i \subset \mathcal{F}$.
3) $\mathbb{P}$ is the probability measure. It is a function $\mathbb{P} : \mathcal{F} \longrightarrow [0,1]$ such that $\mathbb{P}(\Omega) = 1$ and for $\{A_i\}_{i\in\mathbb{N}} \subset \mathcal{F}$ disjoint we have:

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i) \tag{1.1}$$

Definition 1.1.2. *A $\mathbb{P}$-space $(\Omega, \mathcal{F}, \mathbb{P})$ is **complete** if $\mathcal{F}$ contains all sets $A \subset \Omega$ such that there exists $B, C \in \mathcal{F}$, $B \subset A \subset C$, $\mathbb{P}(B) = \mathbb{P}(C)$.*

Definition 1.1.3. *Set $(\Omega, \mathcal{F})$ a measurable space. Let $(E, \mathcal{E})$ be another measurable space. A **E-valued random variable** is a map $X : \Omega \longrightarrow E$ which is $\mathcal{F}/\mathcal{E}$ measurable.*
i.e. $\forall B \subset \mathcal{E}$, $X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\} \subset \mathcal{F}$.

Definition 1.1.4. *If $E = \mathbb{R}$ (or $\mathbb{R}^d$) we take $\mathcal{E} = \mathcal{B}(\mathbb{R})$ to be the **Borel σ-algebra**. It is the smallest σ-algebra containing all open sets of $\mathbb{R}$ (or $\mathbb{R}^d$). A set $B \in \mathcal{B}(\mathbb{R})$ is called a Borel set.*

Definition 1.1.5. *If X is integrable with respect to $\mathbb{P}$, the **expectation of X** is:*

$$\mathbb{E}(X) = \int_\Omega X(\omega) d\mathbb{P}(\omega). \tag{1.2}$$

*If X is furthermore square integrable, the **variance of X** is:*

$$\mathrm{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}(X))^2\right] = \int_\Omega (X(\omega) - \mathbb{E}(X))^2 d\mathbb{P}(\omega). \tag{1.3}$$

Definition 1.1.6. *A random variable $X : \Omega \longrightarrow E$ induces a $\mathbb{P}$-measure on $(E, \mathcal{E})$. For all $B \in \mathcal{E}$:*

$$\mu_X(B) := \mathbb{P}(X^{-1}(B)) = \mathbb{P}\left(\{\omega \in \Omega : X(\omega) \in B\}\right). \tag{1.4}$$

*Here, μ_X is called the **law of X**.*

Remark 1.1.7. *In light of Definition (1.1.5), if $X : \Omega \longrightarrow \mathbb{R}$ is a random variable, $f : \mathbb{R} \longrightarrow \mathbb{R}$ an integrable function:*

$$\mathbb{E}[f(x)] := \int_\Omega f(X(\omega)) d\mathbb{P}(\omega) = \int_E f(x) d\mu_X(x) \tag{1.5}$$

Definition 1.1.8. *For a $\mathbb{P}$-space $(\Omega, \mathcal{F}, \mathbb{P})$ and for any $A, B \in \mathcal{F}$, we say that **A and B are independent** if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.*

Definition 1.1.9. *Given two random variables $X : (\Omega, \mathcal{F}) \longrightarrow (E_1, \mathcal{E}_1)$ and $Y : (\Omega, \mathcal{F}) \longrightarrow (E_2, \mathcal{E}_2)$, let $\boldsymbol{\sigma(X)}$ be the smallest σ-algebra containing all the sets $X^{-1}(B) \subset \mathcal{F}$ where $B \in \mathcal{E}_1$. $\sigma(X)$ is a sub-σ-algebra of $\mathcal{F}$, and we say that **X and Y are independent** if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, for all $A \in \sigma(X)$ and all $B \in \sigma(Y)$. i.e. If and only if $\sigma(X)$ and $\sigma(Y)$ are independent.*

Definition 1.1.10. *Consider $(E_1 \times E_2, \mathcal{E}_1 \otimes \mathcal{E}_2)$ a measure space. We have a product measure $\mu_X \otimes \mu_Y$ where:*

$$\mu_X \otimes \mu_Y(A \times B) = \mu_X(A)\mu_Y(B) \tag{1.6}$$

In particular, if $E_1 = E_2 = \mathbb{R}$ then $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$ and:

$$\begin{aligned}
\mathbb{E}(g(X)h(Y)) &= \int_{\mathbb{R}\times\mathbb{R}} g(x)h(y) d\mu_{X,Y}(x,y) \\
&= \int_{\mathbb{R}\times\mathbb{R}} g(x)h(y) d\mu_X(x) d\mu_Y(y) \\
&= \left(\int_{\mathbb{R}} g(x) d\mu_X(x)\right)\left(\int_{\mathbb{R}} h(y) d\mu_Y(y)\right) = \mathbb{E}(g(X))\mathbb{E}(h(Y))
\end{aligned}$$

Definition 1.1.11. *For a $\mathbb{P}$-space $(\Omega, \mathcal{F}, \mathbb{P})$, let $X : \Omega \longrightarrow \mathbb{R}$ be integrable and $\mathcal{G} \subset \mathcal{F}$ a sub-σ-algebra of $\mathcal{F}$. We define the* ***conditional expectation of* $\mathbf{X}$ *with respect to* $\boldsymbol{\mathcal{G}}$*** *the $\mathcal{G}/\mathcal{B}(\mathbb{R})$-measurable random variable $Y : \Omega \longrightarrow \mathbb{R}$ defined as $Y := \mathbb{E}[X|\mathcal{G}]$, such that for all $A \in \mathcal{G}$:*

$$\int_A Y d\mathbb{P} = \int_A X d\mathbb{P} \tag{1.7}$$

Definition 1.1.12. *Let $Y : (\Omega, \mathcal{F}) \longrightarrow (E, \mathcal{E})$ and $\sigma(Y) \subset \mathcal{F}$ the induced σ-algebra. We define the* ***conditional expectation of* $\mathbf{X}$ *given* $\mathbf{Y}$*** *as the $\sigma(Y)$-measurable random variable $Y := \mathbb{E}[X|\sigma(Y)]$.*

Definition 1.1.13. $\mathbf{L^p(\Omega, \mathcal{F}, \mathbb{P})}$ *is the space of random variables $X : \Omega \longrightarrow \mathbb{R}$ such that:*

$$\int_\Omega |X|^p d\mathbb{P} < \infty. \tag{1.8}$$

We now present a list of properties on the conditional expectation $\mathbb{E}[X|\mathcal{G}]$:

1) $\mathbb{E}\left[\mathbb{E}[X|\mathcal{G}]\right] = \mathbb{E}[X]$ (Tower property).

2) If X is $\mathcal{G}$-measurable $\Longrightarrow X = \mathbb{E}[X|\mathcal{G}]$.

3) $\mathbb{E}[\alpha X + \beta Y|\mathcal{G}] = \alpha\mathbb{E}[X|\mathcal{G}] + \beta\mathbb{E}[Y|\mathcal{G}]$, where $\alpha, \beta \in \mathbb{R}$.

4) If for all $\omega \in \Omega$, $X(\omega) \leq Y(\omega) \Longrightarrow \mathbb{E}[X|\mathcal{G}] \leq \mathbb{E}[Y|\mathcal{G}]$.

5) If $\sigma(X)$ is independent of $\mathcal{G} \Longrightarrow \mathbb{E}[X|\mathcal{G}] = \mathbb{E}[X]$.

6) If Y is $\mathcal{G}$-measurable $\Longrightarrow \mathbb{E}[XY|\mathcal{G}] = Y\mathbb{E}[X|\mathcal{G}]$.

7) If $\phi : \mathbb{R} \longrightarrow \mathbb{R}$ is a convex function and $\phi(X)$ is integrable $\Longrightarrow \phi(\mathbb{E}[X|\mathcal{G}]) \leq \mathbb{E}[\phi(X)|\mathcal{G}]$.

8) If $X \in L^2(\Omega, \mathcal{F}, \mathbb{P})$ and $\mathcal{G} \subset \mathcal{F}$ is a sub-σ-algebra then $Y = \mathbb{E}[X|\mathcal{G}] \in L^2(\Omega, \mathcal{G}, \mathbb{P}) \subset L^2(\Omega, \mathcal{F}, \mathbb{P})$.

1.2 Stochastic Processes

In this section we present the basic notions in Stochastic Processes, where we will study Brownian Motion, martingales, stochastic integrals along with Ito processes. We will then end this section with the necessary tools in Stochastic Calculus.

1.2.1 Stochastic Processes, Filtrations and Stopping Times

Throughout this section, we consider a $\mathbb{P}$-space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition 1.2.1. *A **stochastic process** $\{X_t\}_{\{t\in[0,T]\}}$ in $\mathbb{R}^d$ on $(\Omega,\mathcal{F},\mathbb{P})$ is a collection of $\mathcal{F}/\mathcal{B}(\mathbb{R}^d)$-measurable random variables $X_t:\Omega\longrightarrow\mathbb{R}^d$, for $t\in[0,T]$.*

Definition 1.2.2. *A **sample path** is a function $t\mapsto X_t(\omega)$, where $t\in[0,T]$, $X_t\in\mathbb{R}^d$ and $\omega\in\Omega$.*

Definition 1.2.3. *A **filtration** $\{\mathcal{F}_t\}_t$ is a collection of σ-algebras $\mathcal{F}_t\subset\mathcal{F}$ such that $\mathcal{F}_s\subset\mathcal{F}_t$ for all $s\leq t$.*

Intuitively, a filtration gives us the amount of information known up until time t.

Remark 1.2.4. *A filtration $\{\mathcal{F}_t\}_t$ is right continuous if $\mathcal{F}_t=\bigcap_{s\geq t}\mathcal{F}_s$.*

We now have a Filtered $\mathbb{P}$-space $(\Omega,\mathcal{F},\mathbb{P},\{\mathcal{F}_t\}_t)$. Furthermore, we consider the standard assumption that $(\Omega,\mathcal{F},\mathbb{P})$ is complete, $\{\mathcal{F}_t\}_t$ is right continuous and $\mathcal{F}_t$ contains all $\mathbb{P}$-null sets.

Definition 1.2.5. *We say that a stochastic process on $(\Omega,\mathcal{F},\mathbb{P},\{\mathcal{F}_t\}_t)$ is **$\{\mathcal{F}_t\}_t$-adapted** if X_t is $\mathcal{F}_t$-measurable for all t.*

Definition 1.2.6. *Sometimes a filtration is not given, then we can consider the **natural filtration**:*

$$\{\mathcal{F}_t^X\}_t:=\sigma(X_s,s\leq t). \tag{1.9}$$

Definition 1.2.7. *Let $\{X_t\}_t$, $\{Y_t\}_t$ be two stochastic processes on $(\Omega,\mathcal{F},\mathbb{P},\{\mathcal{F}_t\}_t)$. We say that $\{Y_t\}_t$ is a **version/modification** of $\{X_t\}_t$ if, for all $t\in[0,T]$:*

$$\mathbb{P}(X_t=Y_t)=1. \tag{1.10}$$

*We also say that $\{Y_t\}_t$ is **indistinguishable** from $\{X_t\}_t$ if:*

$$\mathbb{P}\left(X_t=Y_t,\forall t\in[0,T]\right)=1. \tag{1.11}$$

Definition 1.2.8. *We say that $\{X_t\}_t$ is **continuous** if for all $\omega\in\Omega$, $t\mapsto X_t(\omega)$ is a continuous function on $[0,T]$.*

Theorem 1.2.9. *(Kolmogorov's Continuity Theorem) If there exists $\alpha,\beta>0$ such that, for all s,t:*

$$\mathbb{E}\left[|X_t-X_s|^\beta\right]\leq c|t-s|^{1+\alpha}, \tag{1.12}$$

then there exists a continuous process $\{Y_t\}_t$ which is a version of $\{X_t\}_t$. Moreover, every path of Y is γ-Hölder continuous with $\gamma<\alpha/\beta$.

Definition 1.2.10. *We say that $\{X_t\}_t$ is **measurable** if the map $(\omega,t)\mapsto X_t(\omega)$ is $\mathcal{F}\times\mathcal{B}([0,T])$-measurable.*

Definition 1.2.11. *We say that $\{X_t\}_t$ is **progressively measurable** if for all $t\in[0,T]$, the map $(\omega,s)\in\Omega\times[0,T]\mapsto X_s(\omega)$ is $\mathcal{F}_t\times\mathcal{B}([0,T])$-measurable.*

Remark 1.2.12. *If $\{X_t\}_t$ is (right) continuous and adapted, then it is progressively measurable.*

Definition 1.2.13. *A random variable $\tau:\Omega\longrightarrow[0,\infty)$ is called a **stopping time** if $\{\omega\in\Omega:\tau(\omega)\leq t\}\subseteq\mathcal{F}_t$.*

Definition 1.2.14. *Given a stochastic process $\{X_t\}_t$ and a stopping time, we can construct the **stopped process $\mathbf{X_{t\wedge\tau}}$**, which is $\mathcal{F}_t$-measurable,*

$$\mathcal{F}_t=\{A\in\mathcal{F}:A\cap\{\tau\leq t\}\subset\mathcal{F}_t,\forall t\}, \tag{1.13}$$

where $a\wedge b=\min\{a,b\}$.

Example 1.2.15.

$$\tau_N=\inf\{t:|X_t|\geq N\} \tag{1.14}$$

τ_N is a stopping time. The stopped process $X_{t\wedge\tau_N}$ is a bounded process and we can then take a limit to study this process globally.

1.2.2 Brownian Motion in $\mathbb{R}$ and $\mathbb{R}^n$

Definition 1.2.16. *A $\mathbb{R}$-valued adapted stochastic process $\{B_t\}_{t\geq0}$ is a **Brownian Motion** if:*
1) $B_0=0$ a.s.
2) $\forall 0\leq s\leq t$, B_t-B_s is independent of $\mathcal{F}_s$.
3) $\forall 0\leq s\leq t$, $B_t-B_s\sim\mathcal{N}(0,t-s)$.

Again, if the filtration is not given, take the natural filtration,

$$\mathcal{F}_t^B=\sigma(B_s,s\leq t). \tag{1.15}$$

Now if we let $\overline{\mathcal{F}_t^B}$ to be the completion of $\mathcal{F}_t^B$ with the $\mathbb{P}$-null sets,

$$\overline{\mathcal{F}_t^B}=\sigma(\mathcal{F}_t^B\cup N), \tag{1.16}$$

where $N=\{A\in\mathcal{F}:\mathbb{P}(A)=0\}$. One can show that $\{\overline{\mathcal{F}_t^B}\}_t$ is right continuous, and that $\{B_t\}_t$ is still a Brownian Motion with respect to $\{\overline{\mathcal{F}_t^B}\}_t$.

Remark 1.2.17. • *$\{B_t\}_t$ is a Gaussian process, with $B_t \sim \mathcal{N}(0,t)$.*
- *$B_{t+s} - B_s$ is a Brownian Motion with respect to the filtration $\{\mathcal{F}_{t+s}\}_t$.*
- *cB_{t/c^2} is a Brownian Motion with respect to the filtration $\{\mathcal{F}_{t/c^2}\}_t$.*
- *$Z_t = tB_{1/t}\mathbb{1}_{\{t>0\}}$ is a Brownian Motion w.r.t. its own filtration $\mathcal{F}_t = \sigma(Z_s, s \le t)$.*

Proposition 1.2.18.

$$\mathrm{Cov}(B_s, B_t) = \min\{s,t\} \tag{1.17}$$

Proof. Assume that $s \le t$. Then:

$$\begin{aligned}\mathrm{Cov}(B_s,B_t) = \mathbb{E}(B_sB_t) &= \mathbb{E}[B_s(B_t - B_s + B_s)] \\ &= \mathbb{E}[B_s^2] + \mathbb{E}[B_s(B_t - B_s)] \\ &= \mathbb{E}[B_s^2] + \mathbb{E}[B_s]\mathbb{E}[B_t - B_s] = \mathbb{E}[B_s^2] = s.\end{aligned}$$

Hence, by symmetry, for all s,t: $\mathrm{Cov}(B_s,B_t) = s \wedge t = \min\{s,t\}$. □

Remark 1.2.19. *The joint distribution of $B_{t_1}, B_{t_2}, ..., B_{t_n}$ is a multivariate Gaussian $\mathcal{N}(0,\Sigma)$ where:*

$$\Sigma_{ij} = t_i \wedge t_j = \mathrm{Cov}(B_{t_i}, B_{t_j}). \tag{1.18}$$

Existence and Construction of Brownian Motion

1) The first construction is using Kolmogorov's Extension Theorem. The idea is to define Ω as a set of all trajectories of the Brownian Motion itself, i.e. one takes $\Omega = \mathbb{R}^{[0,\infty)}$, hence an element of Ω is a trajectory $\{t \in [0,\infty) : \omega(t)\}$ and $B_t(\omega) = \omega(t)$. The σ-algebra $\mathcal{F}$ is the smallest σ-algebra generated by the cylindrical sets $C = \{\omega \in \Omega : \omega(t_1), ..., \omega(t_n) \in A\}$ with $A \in \mathcal{B}(\mathbb{R}^n)$. On $(\Omega, \mathcal{F})$ one can define a family of finite dimensional probability measures $\pi_{t_1,...,t_n}$ corresponding to the multivariate Gaussian distributions of the Brownian Motion. These distributions are consistent in the sense that for all $A \in \mathcal{B}(\mathbb{R}^{n-1})$, $\pi_{t_1,...,t_{n-1}}(A) = \pi_{t_1,...,t_n}(A \times \mathbb{R})$. Then, Kolmogorov's Extension Theorem guarantees the existence of a unique probability measure on $(\Omega, \mathcal{F})$ whose finite dimensional distributions are $\pi_{t_1,...,t_n}$.

2) A second construction uses a series expansion to define $\{B_t\}_{t\in[0,1]}$.

- For $0 \le t \le 1$ and $Z_i \overset{iid}{\sim} \mathcal{N}(0,1)$, a Fourier series gives us:

$$B_t = Z_0\frac{t}{2\pi} + \frac{\sqrt{2}}{\pi}\sum_{n=1}^{\infty} Z_n \frac{\sin(\pi n t)}{n} \tag{1.19}$$

- For $0 \le t \le 1$ and $Z_{m,k} \overset{iid}{\sim} \mathcal{N}(0,1)$, Lévy-Ciesielski gives us:

$$B_t = \sum_{m=0}^{\infty}\sum_{k\in I(m)} Z_{m,k} S_{m,k}(t) \tag{1.20}$$

where $S_{m,k}(t) = \int_0^t H_{m,k}(s)ds$, also $I(m) = \{j = 0, ..., 2^{m-1} - 1 : 2j + 1\}$ and the Haar basis:

$$H_{m,k}(t) = \begin{cases} 2^{(n-1)/2}, \ \ t \in \left[\frac{k-1}{2^n}, \frac{k}{2^n}\right] \\ -2^{(n-1)/2}, \ \ t \in \left[\frac{k}{2^n}, \frac{k+1}{2^n}\right] \\ 0, \ \ \text{otherwise} \end{cases} \tag{1.21}$$

With these constructions, one takes $\Omega = \mathbb{R}^{\mathbb{N}}$, $\mathcal{F}$ the cylindrical σ-algebra and $\mathbb{P} = \bigotimes_{n=1}^{\infty} \mathcal{N}(0,1)$ so that $(\Omega, \mathcal{F}, \mathbb{P})$ is the probability space of a countable sequence of i.i.d. $\mathcal{N}(0,1)$ random variables and $\omega = (Z_0, Z_1, ...)$.

Regularity of the Sample Paths of $\{B_t\}_t$

- We use Kolmogorov's Continuity Theorem to have:

$$\mathbb{E}[(B_t - B_s)^{2p}] = \mathbb{E}\left[\frac{(B_t - B_s)^{2p}}{(t-s)^p}\right](t-s)^p = \mathbb{E}\left[\left(\frac{B_t - B_s}{\sqrt{t-s}}\right)^{2p}\right](t-s)^p, \tag{1.22}$$

where we have that $\frac{B_t - B_s}{\sqrt{t-s}} \sim \mathcal{N}(0,1)$. Now, for $\xi \sim \mathcal{N}(0,1)$, $\mathbb{E}[|\xi|^{2p}] \leq c_p(t-s)^p$, for some constant c_p. Hence:

$$\mathbb{E}[(B_t - B_s)^{2p}] \leq c_p(t-s)^p. \tag{1.23}$$

Kolmogorov with $\beta = 2p$, $\alpha = p - 1 \Longrightarrow \gamma$-Hölder continuous with $\gamma < \frac{p-1}{2p}$.

Now $\{B_t\}_t$ is almost surely γ-Hölder continuous for $\gamma < \frac{1}{2}$. One can also show that a.s. $B_t(\omega)$ is not γ-Hölder for $\gamma \geq \frac{1}{2}$. Hence, the sample paths $B_t(\omega)$ are a.s. nowhere differentiable.

- The behavior of B_t at $t = 0$ and $t \longrightarrow \infty$ is almost $\sqrt{t}$. For almost every $\omega \in \Omega$, using the law of iterated logarithms, a.s. we have that:

$$\limsup_{t \longrightarrow \infty} \frac{B_t(\omega)}{\sqrt{2t \log\log(t)}} = 1 \tag{1.24}$$

$$\limsup_{t \longrightarrow 0} \frac{B_t(\omega)}{\sqrt{2t \log\log(1/t)}} = 1 \tag{1.25}$$

$$\liminf_{t \longrightarrow \infty} \frac{B_t(\omega)}{\sqrt{2t \log\log(t)}} = -1 \tag{1.26}$$

$$\liminf_{t \longrightarrow 0} \frac{B_t(\omega)}{\sqrt{2t \log\log(1/t)}} = -1 \tag{1.27}$$

Definition 1.2.20. *A $\mathbb{R}^n$-valued adapted stochastic process $\{\overrightarrow{B_t}\}_{t\geq 0}$ is a **Brownian Motion** if:*
1) $\overrightarrow{B_0} = \overrightarrow{0}$ a.s.
2) $\forall 0 \leq s \leq t$, $\overrightarrow{B_t} - \overrightarrow{B_s}$ is independent of $\mathcal{F}_s$.
3) $\forall 0 \leq s \leq t$, $\overrightarrow{B_t} - \overrightarrow{B_s} \sim \mathcal{N}(0, (t-s)\mathbb{1}_{n\times n})$.

$\overrightarrow{B_t} = (B_t^1, ..., B_t^n)$ is a $\mathbb{R}^n$-valued Brownian Motion if and only if B_t^i are independent one-dimensional Brownian Motions.

1.2.3 Total and Quadratic Variations for $\{B_t\}_t$

Let $\pi = \{t_1, ..., t_n\}$ for $0 \leq t_1 < ... < t_n = T$ be a partition of $[0,T]$. Define:

$$|\pi| := \max_{i=0,...,n-1} |t_i - t_{i-1}| \longrightarrow 0. \tag{1.28}$$

Definition 1.2.21. *For $f \in C^1([0,T])$, we define the **Total Variation** to be:*

$$V_\pi^1(f) := \sum_i |f(t_i) - f(t_{i-1})|, \tag{1.29}$$

*and the **Quadratic Variation** as:*

$$V_\pi^2(f) := \sum_i |f(t_i) - f(t_{i-1})|^2. \tag{1.30}$$

Since f is a Lipschitz function we have that:

$$V_\pi^1(f) := \sum_i |f(t_i) - f(t_{i-1})| \leq \max_{[0,T]} |f'| \sum_i |t_i - t_{i-1}| = c < \infty, \tag{1.31}$$

and also that:

$$V_\pi^2(f) := \sum_i |f(t_i) - f(t_{i-1})|^2 \leq \max_i |f(t_i) - f(t_{i-1})| V_\pi^1(f) \xrightarrow{|\pi|\to 0} 0 \tag{1.32}$$

Definition 1.2.22. *For a Brownian Motion $\{B_t\}_t$, we define the **Total Variation** to be:*

$$V_\pi^1(B_t) := \sum_{i=0}^{n-1} |B_{t_i} - B_{t_{i-1}}|, \tag{1.33}$$

*and the **Quadratic Variation** as:*

$$V_\pi^2(B_t) := \sum_{i=0}^{n-1} |B_{t_i} - B_{t_{i-1}}|^2. \tag{1.34}$$

Theorem 1.2.23. *For $\{B_t\}_t$ we have the following:*

$$V_\pi^2(B_t) \xrightarrow{|\pi|\to 0} T \quad \text{in } L^2 \tag{1.35}$$

$$V_\pi^1(B_t) \longrightarrow \infty \quad a.s. \tag{1.36}$$

Proof. • For convergence in L^2 we want to show that, as $|\pi| \to 0$:

$$\mathbb{E}\left[\left(V_\pi^2 - (t-s)\right)^2\right] = 0. \tag{1.37}$$

First we note that we can express $(t-s)$ as:

$$(t-s) = \sum_{i=0}^{m-1} (t_{i+1} - t_i), \tag{1.38}$$

where we will also use the notation $\Delta B_i := B_{t_{i+1}} - B_{t_i}$. Note that $\Delta B_i \sim \mathcal{N}(0, t_{i+1} - t_i)$.
Then we have that:

$$V_\pi^2 - (t-s) = \sum_{k=0}^{m-1} |B_{t_{k+1}} - B_{t_k}|^2 - (t-s) = \sum_{k=0}^{m-1} \left[(\Delta B_k)^2 - (t_{k+1} - t_k)\right]. \tag{1.39}$$

Hence, by independence of the $(\Delta B_k)^2 - (t_{k+1} - t_k)$:

$$\begin{aligned}
\mathbb{E}\left[\left(V_\pi^2 - (t-s)\right)^2\right] &= \sum_{k=0}^{m-1} \mathbb{E}\left[\left((\Delta B_k)^2 - (t_{k+1} - t_k)\right)^2\right] \\
&= \sum_{k=0}^{m-1} \left[\mathbb{E}\left[(\Delta B_k)^4\right] - 2\mathbb{E}\left[(\Delta B_k)^2 (t_{k+1} - t_k)\right] + \mathbb{E}\left[(t_{k+1} - t_k)^2\right]\right] \\
&= \sum_{k=0}^{m-1} \mathbb{E}\left[(\Delta B_k)^4\right] - 2\sum_{k=0}^{m-1} \mathbb{E}\left[(\Delta B_k)^2\right](t_{k+1} - t_k) + \sum_{k=0}^{m-1} \mathbb{E}\left[(t_{k+1} - t_k)^2\right].
\end{aligned}$$

Now we note that we can find the above expectations with the moment generating functions, where for $\Delta B_k \sim \mathcal{N}(0, t_{i+1} - t_i)$, $\mathbb{E}\left[(\Delta B_k)^2\right] = \sigma^2 = (t_{k+1} - t_k)$ and $\mathbb{E}\left[(\Delta B_k)^4\right] = 3\sigma^4 = 3(t_{k+1} - t_k)^2$.
Therefore:

$$\begin{aligned}
\mathbb{E}\left[\left(V_\pi^2 - (t-s)\right)^2\right] &= 3\sum_{k=0}^{m-1} \mathbb{E}\left[(t_{k+1} - t_k)^2\right] - 2\sum_{k=0}^{m-1} \mathbb{E}\left[(t_{k+1} - t_k)^2\right] + \sum_{k=0}^{m-1} \mathbb{E}\left[(t_{k+1} - t_k)^2\right] \\
&= 2\sum_{k=0}^{m-1} \mathbb{E}\left[(t_{k+1} - t_k)^2\right].
\end{aligned}$$

Now we define $|\pi| := \max_{0\le k\le m-1} |t_{k+1} - t_k|$ to obtain:

$$2\sum_{k=0}^{m-1} \mathbb{E}\left[(t_{k+1} - t_k)^2\right] \le 2 \max_{0\le k\le m-1} |t_{k+1} - t_k| \sum_{k=0}^{m-1} \mathbb{E}\left[(t_{k+1} - t_k)\right] = 2|\pi|(t-s). \tag{1.40}$$

Hence, since $\lim_{|\pi|\to 0} 2|\pi|(t-s) = 0$, we conclude that, as $|\pi|0$:

$$\mathbb{E}\left[\left(V_\pi^2 - (t-s)\right)^2\right] = 0 \Longrightarrow \lim_{|\pi|\to 0} V_\pi^2 = (t-s) \quad \text{in } L^2. \tag{1.41}$$

- (Alternative Argument) One may note that:

$$\mathbb{E}\left[\left(V_\pi^2 - (t-s)\right)^2\right] = \sum_{k=0}^{m-1} \mathbb{E}\left[\left((\Delta B_k)^2 - (t_{k+1}-t_k)\right)^2\right] = \sum_{k=0}^{m-1} (t_{k+1}-t_k)^2 \mathbb{E}\left[\left(\frac{(\Delta B_k)^2}{(t_{k+1}-t_k)} - 1\right)^2\right].$$

Now, the quantity $\frac{\Delta B_k}{\sqrt{t_{k+1}-t_k}} \sim \mathcal{N}(0,1)$ and is therefore independent of k. Overall, the expectation in the RHS is a constant and we obtain the following bound:

$$\mathbb{E}\left[\left(V_\pi^2 - (t-s)\right)^2\right] = C\sum_{k=0}^{m-1} (t_{k+1}-t_k)^2 \leq 2C|\pi| \sum_{k=0}^{m-1} (t_{k+1}-t_k) = 2C|\pi|(t-s) \xrightarrow{|\pi|\to 0} 0. \tag{1.42}$$

- For the second statement, we have that:

$$\begin{aligned} V_\pi^2 = \sum_{k=0}^{m-1} |B_{t_{k+1}} - B_{t_k}|^2 &\leq \max_{0\leq k\leq m-1} |B_{t_{k+1}} - B_{t_k}| \sum_{k=0}^{m-1} |B_{t_{k+1}} - B_{t_k}| \\ &= V_\pi^1 \max_{0\leq k\leq m-1} |B_{t_{k+1}} - B_{t_k}|. \end{aligned}$$

Since B is almost surely continuous on $[s,t]$ then it is also uniformly continuous on $[s,t]$. Therefore:

$$\max_{0\leq k\leq m-1} |B_{t_{k+1}} - B_{t_k}| \longrightarrow 0 \Longrightarrow \sum_{k=0}^{m-1} |B_{t_{k+1}} - B_{t_k}| \longrightarrow 0 \quad \text{a.s.} \tag{1.43}$$

This is a contradiction to the first point in the proof, hence for all $t > s$ we necessarily have that:

$$\lim_{|\pi|\to 0} V_\pi^1 = \infty \quad \text{a.s.} \tag{1.44}$$

□

1.2.4 Martingales

Definition 1.2.24. *We say that a stochastic process $\{M_t\}_t$ defined on $(\Omega, \mathcal{F}, \mathbb{P}, \{\mathcal{F}_t\}_t)$ is a* ***martingale*** *with respect to a filtration $\{\mathcal{F}_t\}_t$ if:*
(i) M_t is integrable for all t.
(ii) $\mathbb{E}[M_t|\mathcal{F}_s] = M_s$, for $s \leq t$.

Example 1.2.25. *A Brownian Motion $\{B_t\}_t$ is a martingale:*

$$\mathbb{E}[B_t|\mathcal{F}_s] = \cancelto{0}{\mathbb{E}[B_t - B_s|\mathcal{F}_s]} + \mathbb{E}[B_s|\mathcal{F}_s] = B_s. \tag{1.45}$$

Definition 1.2.26. • *$\{M_t\}_t$ is a* ***super-martingale*** *if $\mathbb{E}[M_t|\mathcal{F}_s] \leq M_s$.*
• *$\{M_t\}_t$ is a* ***sub-martingale*** *if $\mathbb{E}[M_t|\mathcal{F}_s] \geq M_s$.*

Properties of Martingales

• If $\{M_t\}_t$ is a martingale and $\phi : \mathbb{R} \longrightarrow \mathbb{R}$ a convex function and such that $\phi(M_t)$ is integrable for all t, then $Z_t = \phi(M_t)$ is a sub-martingale.
In particular, all the following are sub-martingales: $M_t^+ := \max\{0, M_t\}$, $M_t^- = \min\{0, M_t\}$, $|M_t|^p$ for $p \geq 1$.

• If $\{\mathcal{F}_t\}$ is right continuous and contains all $\mathbb{P}$-null sets, then every martingale $\{M_t\}_t$ has a right continuous version.

• If τ_1 and τ_2 are stopping times with $\tau_1 \leq \tau_2$ a.s. then:

$$\mathbb{E}[M_{\tau_2} | \mathcal{F}_{\tau_1}] = M_{\tau_1}. \tag{1.46}$$

• Let $\widetilde{M_t} = M_{t \wedge \tau}$, for τ a stopping time. Then $\widetilde{M_t}$ is a martingale.

Definition 1.2.27. *Let $\{M_t\}_t$ be square integrable martingale. There exists a unique continuous, adapted, integrable process $\langle M \rangle_t$ called the **Quadratic Variation** of M_t such that $M_t^2 - \langle M \rangle_t$ is a martingale vanishing at $t = 0$.*

Let $\pi = \{0 = t_0 < t_1 < ... < t_n = t\}$ one can show that:

$$\langle M \rangle_t = \lim_{|\pi| \to 0} \sum_{i=0}^{n-1} (M_{t_{i+1}} - M_{t_i})^2 \quad \text{in probability.} \tag{1.47}$$

Example 1.2.28. *For a Brownian Motion $\{B_t\}_t$, $\langle B \rangle_t = t \iff B_t^2 - t$ is a martingale.*

Theorem 1.2.29 (Doob's Martingale Inequality)**.** *If M_t is L^p-integrable, with $p > 1$, then:*

$$\mathbb{E}\left[\sup_{a \leq t \leq b} |M_t|^p\right] \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}[|M_b|^p]. \tag{1.48}$$

Theorem 1.2.30 (Burkholder-Davis-Gundy Inequality)**.** *If M_t is continuous and L^p-integrable, with $p \geq 2$ and k_1, k_2 some constants, then:*

$$k_1 \mathbb{E}\left[\langle M \rangle_b^{p/2}\right] \leq \mathbb{E}\left[\sup_{a \leq t \leq b} |M_t|^p\right] \leq k_2 \mathbb{E}\left[\langle M \rangle_b^{p/2}\right]. \tag{1.49}$$

Remark 1.2.31 (Convergence of Martingales)**.** *If M_t is uniformly bounded in L^p for $p > 1$, i.e. $\exists k > 0$ such that for all t, $\|M_t\|_{L^p(\Omega)} \leq k$. Then there exists $Y \in L^p$ such that $M_t \xrightarrow{t \to \infty} Y$ in L^p and almost surely.*

1.2.5 The Stochastic Integral

Throughout this subsection we consider $\{B_t\}_t$ a Brownian Motion on a filtered $\mathbb{P}$-space $(\Omega, \mathcal{F}, \mathbb{P}, \{\mathcal{F}_t\}_t)$ and another adapted, progressively measurable stochastic process $\{X_t\}_t$ also on $(\Omega, \mathcal{F}, \mathbb{P}, \{\mathcal{F}_t\}_t)$.
The aim is to define an integral of the form:

$$\int_a^b X_t dB_t = \int_a^b X_t(\omega) dB_t(\omega), \quad \text{for all } \omega \in \Omega. \tag{1.50}$$

Of course, if B_t was a.s. differentiable then we could simply define the above integral by:

$$\int_a^b X_t \dot{B}_t dt = \lim_{|\pi| \to 0} \sum_{i=0}^{n} X_{\xi_i}(B_{t_{i+1}} - B_{t_i}), \quad \xi_i \in [t_i, t_{i+1}] \tag{1.51}$$

However, since B_t is a.s. nowhere differentiable, this definition is false.

First, if we consider $X_t = B_t$ such that we want to define $\int_a^b B_t dB_t$ we have two possible constructions. Let $\pi = \{a = t_0 < t_1 < ... < t_n = b\}$ be a partition with:

$$I(\pi, \lambda) := \sum_i B_{\widehat{t_i}}(B_{t_{i+1}} - B_{t_i}), \tag{1.52}$$

where $\widehat{t_i} = \lambda t_{i+1} + (1-\lambda)t_i$, for $\lambda \in [0,1]$. Then:

$$\lim_{|\pi| \to 0} I(\pi, \lambda) = \frac{1}{2}B_b^2 - \frac{1}{2}B_a^2 + \left(\lambda - \frac{1}{2}\right)(b-a) \quad \text{in } L^2. \tag{1.53}$$

- Itô stochastic integral is with $\lambda = 0 \Longrightarrow$ *Non-standard calculus rules*, however $\int_0^t X_s dB_s$ is a martingale.
- Stratonovich stochastic integral is with $\lambda = \frac{1}{2} \Longrightarrow$ *Standard calculus rules.*

Now, similarly to the construction of the Lebesgue integral, we now formalise the stochastic integral first by using simple functions.

Definition 1.2.32. $\mathbf{M^p([a,b])}$ *denotes the class of stochastic processes* $\{X_t\}_{t\in[a,b]}$ *that are progressively measurable such that:*

$$\mathbb{E}\left[\int_a^b |X_s|^p ds\right] < \infty. \tag{1.54}$$

Definition 1.2.33. *We say that* $\{X_t\}_{t\in[a,b]}$ *is a* ***simple process*** *if there exists a partition* $\pi = \{a = t_0 < t_1 < ... < t_n = b\}$ *such that:*

$$X_t(\omega) = \sum_i X_i(\omega) \mathbb{1}_{[t_i, t_{i+1})}(t) \tag{1.55}$$

and X_i *are* $\mathcal{F}_{t_i}$*-adapted.*

Now, for simple functions we define, for $X \in M^2([a,b])$:

$$I = \int_a^b X_t dB_t := \sum_{i=0}^{n-1} X_i (B_{t_{i+1}} - B_{t_i}) = \sum_{i=0}^{n-1} X_i \Delta B_{t_i}. \tag{1.56}$$

Proposition 1.2.34. *For the above definition of I, we have that:*

$$\mathbb{E}[I] = 0 \quad \textit{and}, \quad \mathbb{E}[I^2] = \int_a^b \mathbb{E}(X_t^2) dt. \tag{1.57}$$

Proof. • For the first moment we have that:

$$\mathbb{E}[I] := \mathbb{E}\left[\sum_{i=0}^{n-1} X_i (B_{t_{i+1}} - B_{t_i})\right] = \sum_{i=0}^{n-1} \mathbb{E}\left[\mathbb{E}[X_i \Delta B_{t_i} | \mathcal{F}_{t_i}]\right] = \sum_{i=0}^{n-1} \mathbb{E}\left[X_i \mathbb{E}[\Delta B_{t_i} | \mathcal{F}_{t_i}]\right] = 0, \tag{1.58}$$

where we have used the fact that X_i is $\mathcal{F}_{t_i}$-adapted in the third equality.

• For the second moment we have that:

$$\mathbb{E}[I^2] := \mathbb{E}\left[\sum_{i=0}^{n-1}\sum_{j=0}^{n-1} X_i X_j \Delta B_{t_i} \Delta B_{t_j}\right] = \sum_{i,j} \mathbb{E}\left[X_i X_j \Delta B_{t_i} \Delta B_{t_j}\right]. \tag{1.59}$$

Now, for $j > i$,

$$\mathbb{E}\left[X_i X_j \Delta B_{t_i} \Delta B_{t_j}\right] = \mathbb{E}\left[\mathbb{E}\left[X_i X_j \Delta B_{t_i} \Delta B_{t_j} | \mathcal{F}_{t_j}\right]\right] = 0. \tag{1.60}$$

For $i < j$ we have the same conclusion by symmetry. Then:

$$\begin{aligned}\mathbb{E}[I^2] = \sum_i \mathbb{E}\left[X_i^2 (\Delta B_{t_i})^2\right] = \sum_i \mathbb{E}\left[X_i^2 \mathbb{E}\left[(\Delta B_{t_i})^2 | \mathcal{F}_{t_i}\right]\right] &= \sum_i \mathbb{E}(X_i^2)(t_{i+1} - t_i) \\ &= \int_a^b \mathbb{E}(X_t^2) dt.\end{aligned}$$

□

Theorem 1.2.35 (Itô Isometry)**.**

$$\mathbb{E}\left[\left(\int_a^b X_t dB_t\right)^2\right] = \int_a^b \mathbb{E}(X_t^2) dt. \tag{1.61}$$

Proof. Itô Isometry corresponds to the previous statement in Proposition (1.2.34) $\mathbb{E}[I^2] = \int_a^b \mathbb{E}(X_t^2) dt$.

□

Lemma 1.2.36. *Given $X \in M^2([a,b])$ there exists a sequence $X^{(n)} \in M^2([a,b])$ of simple processes such that:*

$$X^{(n)} \xrightarrow{L^2} X, \quad \textit{i.e.} \lim_{n\to\infty} \mathbb{E}\left[\int_a^b \left(X_t - X_t^{(n)}\right)^2 dt\right] = 0. \tag{1.62}$$

Remark 1.2.37. *We have that* $\left(X^{(n)} - X^{(m)}\right)$ *is a simple process. Using Itô Isometry we have that:*

$$\mathbb{E}\left[\left(\int_a^b \left(X_t^{(n)} - X_t^{(m)}\right) dB_t\right)^2\right] = \mathbb{E}\left[\left(\int_a^b X_t^{(n)} dB_t - \int_a^b X_t^{(m)} dB_t\right)^2\right]$$
$$= \int_a^b \mathbb{E}\left[\left(X_t^{(n)} - X_t^{(m)}\right)^2\right] dt \xrightarrow{n,m\to\infty} 0.$$

Since we have Cauchy sequences, it implies convergence.
The sequence $I^{(n)} := \int_a^b X_t^{(n)} dB_t \in L^2(\Omega)$ *has a limit:*

$$\int_a^b X_t dB_t := \lim_{n\to\infty} \int_a^b X_t^{(n)} dB_t. \tag{1.63}$$

This limit does not depend on the approximating sequence.

Properties of the Stochastic Integral $\mathbf{I = \int_a^b X_t dB_t}$

- $\mathbb{E}[I] = 0$ and $\mathbb{E}[I^2] = \int_a^b \mathbb{E}(X_t^2) dt$
- (Linearity) For all $X_t, Y_t \in M^2([a,b])$ and $\alpha, \beta \in \mathbb{R}$: $\int_a^b (\alpha X_t + \beta Y_t) dB_t = \alpha \int_a^b X_t dB_t + \beta \int_a^b Y_t dB_t$.
- For $c \in [a,b]$: $\int_a^b X_t dB_t = \int_a^c X_t dB_t + \int_c^b X_t dB_t$.
- $\mathbb{E}\left[\int_a^b X_t dB_t | \mathcal{F}_a\right] = 0$ and $\mathbb{E}\left[\left(\int_a^b X_t dB_t\right)^2 | \mathcal{F}_a\right] = \int_a^b \mathbb{E}\left[X_t^2 | \mathcal{F}_a\right] dt$.

Indefinite Integral $\mathbf{I_t := \int_0^t X_s dB_s}$
I_t is a stochastic process, $\mathcal{F}_t$-measurable and square integrable.

- I_t is a martingale:

$$\mathbb{E}[I_t | \mathcal{F}_s] = \mathbb{E}\left[\int_0^s X_p dB_p + \int_s^t X_p dB_p \Big| \mathcal{F}_s\right] = \int_0^s X_p dB_p = I_s, \quad \text{for } s \le t. \tag{1.64}$$

- The quadratic variation of I_t is:

$$\langle I \rangle_t = \int_0^t X_s^2 ds. \tag{1.65}$$

- Doob's Martingale Inequality gives us:

$$\mathbb{E}\left[\sup_{0\le t\le T} |I_t|^2\right] = \mathbb{E}\left[\sup_{0\le t\le T} \left(\int_0^t X_s dB_s\right)^2\right] \le 4\mathbb{E}\left[\left(\int_0^T X_s dB_s\right)^2\right] = 4\int_0^T \mathbb{E}(X_s^2) ds, \tag{1.66}$$

where the last equality comes from Itô Isometry.

1.2.6 Itô Processes

In this section, we consider a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P}, \{\mathcal{F}_t\}_t)$ and $\{B_t\}_t$ a Brownian Motion with respect to $\{\mathcal{F}_t\}_t$.

Definition 1.2.38. *Let $f \in M^1([0,T])$ and $g \in M^2([0,T])$. An **Itô process** $\{X_t\}_t$ is a process of the form:*

$$X_t = X_0 + \underbrace{\int_0^t f_s ds}_{J_t} + \underbrace{\int_0^t g_s dB_s}_{I_t}. \tag{1.67}$$

- I_t is a martingale with bounded quadratic variation and infinite total variation.
- J_t is a *standard integral* with bounded total variation and zero quadratic variation.
- Following Definition (1.2.22) and considering a partition $\pi = \{0 = t_0 < t_1 < ... < t_n = t\}$ we have that the total variation for J is bounded:

$$V_\pi^1(J) = \sum_{j=0}^{n-1} |J_{t_{j+1}} - J_{t_j}| = \sum_{j=0}^{n-1} \left| \int_{t_j}^{t_{j+1}} f_s ds \right| \leq \sum_{j=0}^{n-1} \int_{t_j}^{t_{j+1}} |f_s| ds = \int_0^t |f_s| ds < \infty. \tag{1.68}$$

Proposition 1.2.39. *The representation of an Itô Process is unique.*

Proof. Suppose otherwise and that there exists another Itô Process such that:

$$X_t = X_0 + \int_0^t f_s ds + \int_0^t g_s dB_s = X_0 + \int_0^t \tilde{f}_s ds + \int_0^t \tilde{g}_s dB_s. \tag{1.69}$$

Then rearranging we obtain:

$$\int_0^t (f_s - \tilde{f}_s) ds = \int_0^t (\tilde{g}_s - g_s) dB_s. \tag{1.70}$$

Therefore we have:

$$\underbrace{J_t - \tilde{J}_t}_{\text{Bounded variation + zero quadratic variation}} = \underbrace{\tilde{I}_t - J_t}_{\text{Zero total variation + bounded quadratic variation}}. \tag{1.71}$$

Hence we have uniqueness:

$$\begin{cases} J_t = \tilde{J}_t, & \forall t \\ I_t = \tilde{I}_t, & \forall t \end{cases} \implies \begin{cases} f_t = \tilde{f}_t, & \text{a.e. in } t \\ g_t = \tilde{g}_t, & \text{a.e. in } t \end{cases}. \tag{1.72}$$

□

Definition 1.2.40. *Similarly to Definition (1.2.27), the **Quadratic Variation** of $\{X_t\}_t$ is:*

$$\langle X \rangle_t := \langle I \rangle_t = \int_0^t (g_s)^2 ds. \tag{1.73}$$

And again following from Equation (1.47), for $\pi = \{0 = t_0 < t_1 < ... < t_n = t\}$ one can show that:

$$\langle X \rangle_t = \lim_{|\pi| \to 0} \sum_{j=0}^{n-1} (X_{t_{j+1}} - X_{t_j})^2 \quad \text{in probability.} \tag{1.74}$$

Definition 1.2.41. *Consider the following process, for $i \in \{1,2\}$:*

$$X_t^{(i)} = X_0^{(i)} + \int_0^t f_s^{(i)} ds + \int_0^t g_s^{(i)} dB_s. \tag{1.75}$$

*We then define the **Cross-Variation** of $X^{(1)}$ and $X^{(2)}$ as:*

$$\langle X^{(1)}, X^{(2)} \rangle_t := \int_0^t g_s^{(1)} g_s^{(2)} ds. \tag{1.76}$$

Definition 1.2.42. *We define the **Stochastic Differential Equation** of $\{X_t\}_t$ as:*

$$dX_t = f_t dt + g_t dB_t, \tag{1.77}$$

which is an equivalent notation to the integral form of (1.67).

1.2.7 Chain Rule for Itô Processes

In this section, we present the heuristics behind the Itô formulae that we will introduce in the next subsection.

Deterministic Setting

In this setting we consider $dX_t = f_t dt$ for $f \in L^1([0,T])$. Let $u : \mathbb{R} \longrightarrow \mathbb{R}$ be a smooth function such that $Y_t = u(X_t)$. Then:

$$dY_t = u'(X_t) dX_t = u'(X_t) f_t dt \Longrightarrow Y_t = u(X_0) + \int_0^t u'(X_s) f_s ds. \tag{1.78}$$

Indeed, writing Y_t with a telescoping sum and Taylor expanding to order 2 we have:

$$\begin{aligned} Y_t = u(X_t) &= u(X_0) + \sum_{j=0}^{n-1} \Big(u(X_{t_{j+1}}) - u(X_{t_j}) \Big) \\ &= u(X_0) + \sum_{j=0}^{n-1} u'(X_{t_j})(X_{t_{j+1}} - X_{t_j}) + \frac{1}{2} \sum_{j=0}^{n-1} u''(\xi_j)(X_{t_{j+1}} - X_{t_j})^2, \quad \xi_j \in [X_{t_j}, X_{t_{j+1}}] \\ &= u(X_0) + \sum_{j=0}^{n-1} \int_{t_j}^{t_{j+1}} u'(X_{t_j}) f_t dt + \frac{1}{2} \sum_{j=0}^{n-1} u''(\xi_j)(X_{t_{j+1}} - X_{t_j})^2. \end{aligned}$$

Now note that we have the following two convergences,

$$\sum_{j=0}^{n-1} \int_{t_j}^{t_{j+1}} u'(X_{t_j}) f_t dt \longrightarrow \int_0^t u'(X_s) f_s ds, \tag{1.79}$$

and,

$$\frac{1}{2} \sum_{j=0}^{n-1} u''(\xi_j)(X_{t_{j+1}} - X_{t_j})^2 \leq \frac{1}{2} \max_j |X_{t_{j+1}} - X_{t_j}| \sum_{j=0}^{n-1} |u''(\xi_j)||X_{t_{j+1}} - X_{t_j}| \longrightarrow 0. \tag{1.80}$$

Hence we conclude that, as expected:

$$Y_t \longrightarrow u(X_0) + \int_0^t u'(X_s) f_s ds. \tag{1.81}$$

Stochastic Setting

For a stochastic differential equation,

$$dX_t = f_t dt + g_t dB_t, \quad Y_t = u(X_t) \tag{1.82}$$

First note that in a stochastic setting, we can write $(X_{t_{j+1}} - X_{t_j})$ as:

$$(X_{t_{j+1}} - X_{t_j}) = \int_{t_j}^{t_{j+1}} (f_t dt + g_t dB_t). \tag{1.83}$$

Now, similarly to the deterministic setting we Taylor expand to order 2 to find:

$$\begin{aligned} Y_t = u(X_t) &= u(X_0) + \sum_{j=0}^{n-1} \Big(u(X_{t_{j+1}}) - u(X_{t_j}) \Big) \\ &= u(X_0) + \sum_{j=0}^{n-1} u'(X_{t_j})(X_{t_{j+1}} - X_{t_j}) + \frac{1}{2} \sum_{j=0}^{n-1} u''(\xi_j)(X_{t_{j+1}} - X_{t_j})^2 \\ &= Y_0 + \sum_{j=0}^{n-1} \left[\int_{t_j}^{t_{j+1}} u'(X_{t_j}) f_t dt + \int_{t_j}^{t_{j+1}} u'(X_{t_j}) g_t dB_t \right] + \underbrace{\frac{1}{2} \sum_{j=0}^{n-1} u''(\xi_j)(X_{t_{j+1}} - X_{t_j})^2}_{\equiv A}. \end{aligned}$$

Now note that we have the following three convergences,

$$\bullet \sum_{j=0}^{n-1} \int_{t_j}^{t_{j+1}} u'(X_{t_j}) f_t dt \longrightarrow \int_0^t u'(X_s) f_s ds, \tag{1.84}$$

$$\bullet \sum_{j=0}^{n-1} \int_{t_j}^{t_{j+1}} u'(X_{t_j}) g_t dB_t \longrightarrow \int_0^t u'(X_s) g_s dB_s, \tag{1.85}$$

$$\bullet A = \frac{1}{2} \sum_j u''(\xi_j)(dX_t)^2 \longrightarrow \frac{1}{2} \int_0^t u''(X_s)(g_s)^2 ds. \tag{1.86}$$

This yields the Itô Process:

$$Y_t \longrightarrow Y_0 + \int_0^t \left(u'(X_s) f_s + \frac{1}{2} u''(X_s)(g_s)^2 \right) ds + \int_0^t u'(X_s) g_s dB_s. \tag{1.87}$$

Hence as long as $(u'(x)f + \frac{1}{2}u''(x)g) \in M^1([0,T])$ and $u'(x)g \in M^2([0,T])$, then Y_t is an Itô Process with:

$$dY_t = \left(u'(X_t) f_t + \frac{1}{2} u''(X_t)(g_t)^2 \right) dt + u'(X_t) g_t dB_t. \tag{1.88}$$

One can reduce this formula in a more compact form where:

$$\begin{aligned} dY_t = d(u(X_t)) &= u'(X_t) dX_t + \frac{1}{2} u''(X_t)(dX_t)^2 \\ &= u'(X_t) f_t dt + u'(X_t) g_t dB_t + \frac{1}{2} u''(X_t) \left[f_t^2 \underbrace{\cancelto{0}{dt^2}}_{\text{H.O.T.}} + f_t g_t \underbrace{\cancelto{0}{dt dB_t}}_{\text{H.O.T.}} + g_t^2 \underbrace{dB_t^2}_{=dt} \right], \end{aligned}$$

where the letters H.O.T. are for *Higher Order Term*, which implies that dt^2 and $dtdB_t$ are negligible in the expansion. Therefore, we have the following formula in compact form:

$$dY_t = u'(X_t)f_t dt + u'(X_t)g_t dB_t + \frac{1}{2}u''(X_t)g_t^2 dt. \tag{1.89}$$

Or equivalently:

$$dY_t = du(X_t) = u'(X_t)dX_t + \frac{1}{2}u''(X_t)d\langle X\rangle_t. \tag{1.90}$$

1.2.8 Itô Formulae

In light of the previous subsection, we are now ready to give the theorems for the Itô formula in 1-dimension, the Itô formula in 1-dimension with multiple Brownian Motions and finally the Itô formula in d-dimensions.

Itô Formula in 1-Dimension

Theorem 1.2.43 (Itô Formula - $d = 1$)**.** *Consider two functions $f \in M^1([0,T])$ and $g \in M^2([0,T])$ such that we have the following process:*

$$dX_t = f_t dt + g_t dB_t. \tag{1.91}$$

Given $u(x,t) : \mathbb{R}\times\mathbb{R}_+ \longrightarrow \mathbb{R}$ with continuous partial derivatives $\partial_t u$, $\partial_x u$ and $\partial_{xx}u$, we have that $Y_t = u(X_t,t)$ is an Itô Process with differential:

$$\begin{aligned} dY_t = du(X_t,t) &= \frac{\partial}{\partial t}u(X_t,t)dt + \frac{\partial}{\partial x}u(X_t,t)dX_t + \frac{1}{2}\frac{\partial^2}{\partial x^2}u(X_t,t)d\langle X\rangle_t \\ &= \left(\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x}f + \frac{1}{2}\frac{\partial^2 u}{\partial x^2}g^2\right)dt + \left(\frac{\partial u}{\partial x}g\right)dB_t, \end{aligned}$$

where we have used the fact that $d\langle X\rangle_t = g_t^2 dt$.

Example 1.2.44. *We want to find dB_t^2. Let $X_t = B_t \Longrightarrow dX_t = dB_t$. Then $u(x,t) = x^2$ and by Itô formula we have:*

$$dB_t^2 = du(X_t) = 2X_t dX_t + \frac{1}{2}(2)d\langle X\rangle_t = dt + 2B_t dB_t. \tag{1.92}$$

Itô Formula in 1-Dimension with Multiple Brownian Motions

Theorem 1.2.45 (Itô Formula - $d=1$ with Multiple Brownian Motions)**.** *Consider two functions $f \in M^1([0,T])$, $g \in M^2([0,T])$ and integers $k \in \{1,...,m\}$ such that we have the following process:*

$$dX_t = f_t dt + \sum_{k=1}^{m} g_t^{(k)} dB_t^{(k)}, \tag{1.93}$$

with $\overrightarrow{B_t} = \left(B_t^{(1)},...,B_t^{(m)}\right)$ an m-dimensional Brownian Motion ($B_t^{(i)}$ are mutually independent). Further let $u(x,t) : \mathbb{R} \times \mathbb{R}_+ \longrightarrow \mathbb{R}$ with continuous partial derivatives $\partial_t u$, $\partial_x u$ and $\partial_{xx} u$, we have that $Y_t = u(X_t,t)$ is an Itô Process with differential:

$$\begin{aligned} dY_t &= \frac{\partial}{\partial t}u(X_t,t)dt + \frac{\partial}{\partial x}u(X_t,t)dX_t + \frac{1}{2}\frac{\partial^2}{\partial x^2}u(X_t,t)d\langle X\rangle_t \\ &= \left(\partial_t u + \partial_x u f_t + \frac{1}{2}\partial_{xx} u \sum_{k=1}^{m}\left(g_t^{(k)}\right)^2\right)dt + \sum_{k=1}^{m}\partial_x u g_t^{(k)} dB_t^{(k)}. \end{aligned}$$

We justify how to find $d\langle X\rangle_t$ in the above Theorem. First we note that the process in consideration is of the form:

$$X_t = X_0 + J_t + \sum_{k=1}^{m} I_t^{(k)}, \tag{1.94}$$

where $J_t := \int_0^t f_s ds$ and $I_t^{(k)} := \int_0^t g_s dB_s^{(k)}$. Furthermore we can remark that for any $k \neq k'$, $\mathbb{E}\left[I_t^{(k)} I_t^{(k')}\right] = 0$. Therefore, aiming to have an expression for $\langle X\rangle_t$ we find:

$$\langle X\rangle_t = \left\langle \sum_{k=1}^{m} I_t^{(k)} \right\rangle = \sum_{k=1}^{m}\left\langle I^{(k)}\right\rangle_t = \sum_{k=1}^{m}\int_0^t \left(g_s^{(k)}\right)^2 ds = \int_0^t \sum_{k=1}^{m}\left(g_s^{(k)}\right)^2 ds, \quad \text{as required.} \tag{1.95}$$

Itô Formula in d-Dimensions

Theorem 1.2.46 (Itô Formula in Dimension d)**.** *Consider two functions $f^{(i)} \in M^1([0,T])$, $g^{(i,k)} \in M^2([0,T])$ and integers $k \in \{1,...,m\}$, $i \in \{1,...,d\}$ such that we have the following process:*

$$dX_t^{(i)} = f_t^{(i)} dt + \sum_{k=1}^{m} g_t^{(i,k)} dB_t^{(k)}. \tag{1.96}$$

In vector notation it is given by $d\overrightarrow{X_t} = \overrightarrow{f_t}dt + G_t d\overrightarrow{B_t}$, where $\overrightarrow{X_t} = \left(X_t^{(1)},...,X_t^{(d)}\right)^T$, $\overrightarrow{B_t} = \left(B_t^{(1)},...,B_t^{(m)}\right)$, $\overrightarrow{f_t} = \left(f_t^{(1)},...,f_t^{(d)}\right)$ and $G_t = \left(g_t^{(i,k)}\right)_{i,k} \in \mathbb{R}^{d\times m}$.
Further let $u(x_1,...,x_d,t) : \mathbb{R}^d \times \mathbb{R}_+ \longrightarrow \mathbb{R}$ with continuous partial derivatives $\partial_t u$, $\partial_{x_i} u$ and $\partial_{x_i x_j} u$, we have that $Y_t = u\left(X_t^{(1)},...,X_t^{(d)},t\right) = u\left(\overrightarrow{X_t},t\right)$ is a 1-dimensional Itô Process with differential:

$$\begin{aligned} dY_t &= \frac{\partial}{\partial t}u\left(\overrightarrow{X_t},t\right)dt + \sum_{i=1}^{d}\frac{\partial}{\partial x_i}u\left(\overrightarrow{X_t},t\right)dX_t^{(i)} + \frac{1}{2}\sum_{i,j=1}^{d}\frac{\partial^2}{\partial x_i \partial x_j}u\left(\overrightarrow{X_t},t\right)d\langle X^{(i)},X^{(j)}\rangle_t \\ &= \left(\partial_t u + \sum_{i=1}^{d}\partial_{x_i} u f_t^{(i)} + \frac{1}{2}\sum_{k=1}^{m}\sum_{i,j=1}^{d}\partial_{x_i x_j} u g_t^{(i,k)} g_t^{(j,k)}\right)dt + \sum_{k=1}^{m}\sum_{i=1}^{d}\partial_{x_i} u g_t^{(i,k)} dB_t^{(k)}. \end{aligned}$$

We again justify how to find $d\langle X^{(i)}, X^{(j)}\rangle_t$ in the above Theorem:

$$\begin{aligned} d\langle X^{(i)}, X^{(j)}\rangle_t &= d\left\langle \sum_{k=1}^{m} g_t^{(i,k)} dB_t^{(k)}, \sum_{k'=1}^{m} g_t^{(j,k')} dB_t^{(k')} \right\rangle_t \\ &= \sum_{k=1}^{m} d\left\langle g_t^{(i,k)} dB_t^{(k)}, g_t^{(j,k)} dB_t^{(k)} \right\rangle_t \\ &= \sum_{k=1}^{m} g_t^{(i,k)} g_t^{(j,k)} dt, \quad \text{as required.} \end{aligned}$$

Definition 1.2.47. *We define* $(A : B)$ *to be:*

$$(A : B) := \sum_{i,j} A_{ij} B_{ij} = \text{Trace}(AB^T) = \text{Trace}(BA^T). \tag{1.97}$$

Using the notation defined above, we can write the differential of the Theorem in the following compact form:

$$dY_t := du\left(\overrightarrow{X}_t, t\right) = \left(\partial_t u + \left(\overrightarrow{f}_t \cdot \nabla\right) u + \frac{1}{2} G_t G_t^T : \nabla^2 u\right) dt + \sum_{k=1}^{m} \left(g_t^{(i,k)} \cdot \nabla\right) u dB_t^{(k)}. \tag{1.98}$$

Chapter 2

Stochastic Differential Equations (SDEs)

We now begin the theory of Stochastic differential equations. We will first introduce some different types of solutions, their existence and their uniqueness, followed by a list of several properties. We will then define a Generator associated to an SDE followed by the definition of a Generator of semigroup.

2.1 Definitions and Important SDEs

In this section we give the definitions of an SDE, the drift and the diffusion. We then present the SDE formulations for an Ornstein-Uhlenbeck process and for a Geometric Brownian Motion.

Definition 2.1.1. *Let $\overrightarrow{b} : \mathbb{R}^d \times \mathbb{R}_+ \longrightarrow \mathbb{R}^d$ and $\sigma : \mathbb{R}^d \times \mathbb{R}_+ \longrightarrow \mathbb{R}^{d\times m}$ to be deterministic maps. Furthermore let $\overrightarrow{\eta}$ be an $\mathcal{F}_0$-measurable random variable. A **Stochastic Differential Equation (SDE)** is an equation of the form:*

$$\begin{cases} d\overrightarrow{X_t} = \overrightarrow{b}\left(\overrightarrow{X_t}, t\right) dt + \sigma\left(\overrightarrow{X_t}, t\right) d\overrightarrow{B_t} \\ \overrightarrow{X_0} = \overrightarrow{\eta} \end{cases} \tag{2.1}$$

In integral form we have:

$$\overrightarrow{X_t} = \overrightarrow{X_0} + \int_0^t \overrightarrow{b}\left(\overrightarrow{X_s}, s\right) ds + \int_0^t \sigma\left(\overrightarrow{X_s}, s\right) d\overrightarrow{B_s}. \tag{2.2}$$

Component-wise we have:

$$dX_t^{(i)} = b^{(i)}\left(\overrightarrow{X_t}, t\right) dt + \sum_{k=1}^{m} \sigma^{(i,k)}\left(\overrightarrow{X_t}, t\right) dB_t^{(k)}. \tag{2.3}$$

Definition 2.1.2. *In a SDE, $\overrightarrow{b}\left(\overrightarrow{X_t}, t\right)$ is called the **drift**.*

Definition 2.1.3. *In a SDE, $\sigma\left(\overrightarrow{X_t}, t\right)$ is called the **diffusion**.*

Example 2.1.4 (Ornstein-Uhlenbeck Process)**.** *Consider the Ornstein-Uhlenbeck Process given by:*

$$\begin{cases} dX_t = -\lambda X_t dt + \sigma dB_t, \ \ t \geq 0 \\ X_0 = \eta \in \mathbb{R} \end{cases} \tag{2.4}$$

In the deterministic case $\sigma = 0$, using the method of integrating factors we have that $X_t = \eta e^{-\lambda t}$. Now we can suggest the transformation $Y_t = e^{\lambda t} X_t$. Then using Itô Formula we get:

$$\begin{aligned} dY_t = \lambda e^{\lambda t} X_t dt + e^{\lambda t} dX_t &= \lambda e^{\lambda t} X_t dt + e^{\lambda t}(-\lambda X_t dt + \sigma dB_t) \\ &= \sigma e^{\lambda t} dB_t. \end{aligned}$$

Hence we obtain the following Gaussian Process: $Y_t = \eta + \int_0^t e^{\lambda s} \sigma dB_s$.
And using the transformation we get $X_t = e^{-\lambda t}\eta + e^{-\lambda t}\int_0^t e^{\lambda s}\sigma dB_s$. We note that if η is Gaussian then X_t is a Gaussian Process.

Example 2.1.5 (Geometric Brownian Motion)**.** *Consider the Geometric Brownian Motion given by:*

$$\begin{cases} dX_t = \mu X_t dt + \sigma X_t dB_t, \ \ t \geq 0 \\ X_0 = x > 0 \end{cases} \tag{2.5}$$

Looking at the deterministic case $\sigma = 0$, we can suggest the transformation $Y_t = \log(X_t)$. Then using Itô Formula we get:

$$\begin{aligned} dY_t = \frac{1}{X_t} dX_t + \frac{1}{2}\left(-\frac{1}{X_t^2}\right) d\langle X \rangle_t &= \frac{1}{X_t}(\mu X_t dt + \sigma X_t dB_t) - \frac{1}{2X_t^2}\sigma^2 X_t^2 dt \\ &= \left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma dB_t. \end{aligned}$$

Hence we obtain the following process:

$$Y_t = Y_0 + \left(\mu - \frac{\sigma^2}{2}\right) t + \sigma B_t = \log(x) + \left(\mu - \frac{\sigma^2}{2}\right) t + \sigma B_t.$$

We note that $Y_t \sim \mathcal{N}\left(\log(x) + \left(\mu - \frac{\sigma^2}{2}\right) t, \sigma^2\right)$. Now using the transformation we get $X_t = e^{Y_t} = x e^{\left(\mu - \frac{\sigma^2}{2}\right) t + \sigma B_t}$. We remark that X_t follows a log-normal distribution.

2.2 Types, Existence, Uniqueness and Properties of Solutions

In this section we define the notions of strong and weak solutions of SDEs along with statements of the uniqueness of each of them. We will then end on three sets of assumptions that will be useful for a theorem on the existence of solutions, and for several properties.
In what follows, for a matrix A, we will use the Frobenius Norm defined by $\|A\|_F := \sqrt{\text{Trace}(A^* A)} = \sqrt{\sum_i \sum_j |a_{ij}|^2}$.

Definition 2.2.1. *We say that a process $\{X_t\}_t$ is a **strong solution** if:*
1) X_t is $\mathcal{F}_t$-adapted and continuous.
2) $\{b(X_t,t)\}_t \in M^1([0,T])$ and $\{\sigma(X_t,t)\}_t \in M^2([0,T])$.
3) The SDE in integral form (2.2) holds a.s. for all $t \geq 0$.

Definition 2.2.2. *We call a **weak solution** a triple,*

$$((\Omega,\mathcal{F},\mathbb{P},\{\mathcal{F}_t\}_t),\{B_t\}_t,\{X_t\}_t), \tag{2.6}$$

such that X_t is continuous, $\mathcal{F}_t$-adapted and (2.2) is satisfied a.s. with $X_0 \sim \eta$.

Definition 2.2.3. *A strong solution is **strongly/pathwise unique** if any other strong solution $\{\widetilde{X}_t\}_t$ is indistinguishable from $\{X_t\}_t$:*

$$\mathbb{P}\left(\widetilde{X_t} = X_t, \forall t\right) = 1, \tag{2.7}$$

i.e. $\widetilde{X_t}(\omega) = X_t(\omega)$, for all $t \geq 0$ and for almost every ω.

Definition 2.2.4. *A weak solution is **weakly unique** if any other weak solution $\{\widetilde{X_t}\}_t$ (defined possibly on a different Brownian Motion) has the same Law as $\{X_t\}_t$:*

$$\forall t_1,...,t_m \in [0,T] : \left(\widetilde{X_{t_1}},...,\widetilde{X_{t_m}}\right) \sim (X_{t_1},...,X_{t_m}). \tag{2.8}$$

Assumptions (A)

- There exists $L > 0$ such that global Lipschitz conditions hold:

$$|b(x,t)-b(y,t)| \leq L|x-y| \quad \text{and,} \quad \|\sigma(x,t)-\sigma(y,t)\|_F \leq L|x-y|. \tag{2.9}$$

- For $M > 0$ we have a Linear Growth Bound (L.G.B.):

$$|b(x,t)| + \|\sigma(x,t)\|_F \leq M(1+|x|). \tag{2.10}$$

- We have $\eta \in L^2(\Omega)$.

Assumptions (B)

- For all N, there exists $L_N > 0$ such that for all $|x|,|y| \leq N$ a local Lipschitz condition holds:

$$|b(x,t)-b(y,t)| + \|\sigma(x,t)-\sigma(y,t)\|_F \leq L_N|x-y|. \tag{2.11}$$

- For $M > 0$ we have a Linear Growth Bound (L.G.B.):

$$|b(x,t)| + \|\sigma(x,t)\|_F \leq M(1+|x|). \tag{2.12}$$

Assumptions (C)

- For all N, there exists $L_N > 0$ such that for all $|x|, |y| \leq N$ a local Lipschitz condition holds:

$$|b(x,t) - b(y,t)| + \|\sigma(x,t) - \sigma(y,t)\|_F \leq L_N |x-y|. \tag{2.13}$$

- For $M > 0$ we have a monotonicity condition:

$$x^T b(x,t) + \|\sigma(x,t)\|_F^2 \leq M(1+|x|^2). \tag{2.14}$$

Theorem 2.2.5. *Under Assumptions (A), there exists a unique strong solution $\{X_t\}_t$ and $X \in M^2([0,T])$.*

Proposition 2.2.6 (L^p Bound)**.** *Under Assumptions (A) or (B) or (C) and $\eta \in L^p(\Omega)$, for some $p \geq 2$ and some constant $C(T,M,p)$ we have:*

$$\mathbb{E}\left[\sup_{0\leq t\leq T} |X_t|^p\right] \leq C(T,M,p)(1+\mathbb{E}[|\eta|^p]). \tag{2.15}$$

We present the proof in the case $p = 2$, where for higher values of p extra details and inequalities are required. Proofs for $p > 2$ are left to the interest of the reader.

Proof. (For $p = 2$) Consider a process $X_t = \eta + \int_0^t b(X_s,s)ds + \int_0^t \sigma(X_s,s)dB_s$. We first use Young's Inequality to find:

$$\begin{aligned}
\mathbb{E}\left[\sup_{0\leq t\leq T} |X_t|^2\right] &= \mathbb{E}\left[\sup_{0\leq t\leq T} \left|\eta + \int_0^t b(X_s,s)ds + \int_0^t \sigma(X_s,s)dB_s\right|^2\right] \\
&\leq \mathbb{E}\left[\sup_{0\leq t\leq T} \left(3|\eta|^2 + 3\left|\int_0^t b(X_s,s)ds\right|^2 + 3\left|\int_0^t \sigma(X_s,s)dB_s\right|^2\right)\right].
\end{aligned}$$

Now, distributing the supremum to each term we find for the drift term:

$$\begin{aligned}
\mathbb{E}\left[\sup_{0\leq t\leq T} 3\left|\int_0^t b(X_s,s)ds\right|^2\right] &\leq \mathbb{E}\left[3T\int_0^T |b(X_s,s)|^2 ds\right], \quad \text{by Cauchy-Schwarz Inequality,} \\
&\leq \mathbb{E}\left[3T\int_0^T M^2(1+|X_s|)^2 ds\right] \\
&\leq \mathbb{E}\left[3T\int_0^T M^2(2+2|X_s|^2) ds\right] \\
&\leq C_1 + C_2 \int_0^T \underbrace{\mathbb{E}\left[\sup_{0\leq u\leq T} |X_u|^2\right]}_{\equiv E_s} ds,
\end{aligned}$$

for some constants C_1 and C_2. Continuing with the diffusion term, and assuming without proof

that $\left|\int_0^t \sigma(X_s,s)dB_s\right|^2$ is a sub-martingale we have that:

$$\begin{aligned}\mathbb{E}\left[\sup_{0\le t\le T}\left|\int_0^t \sigma(X_s,s)dB_s\right|^2\right] &\le 4\mathbb{E}\left[\left|\int_0^T \sigma(X_s,s)dB_s\right|^2\right], \quad \text{by Doob's Inequality,}\\ &= 4\int_0^T \mathbb{E}\left[\|\sigma(X_s,s)\|^2\right]ds, \quad \text{by Itô Isometry,}\\ &\le 4\int_0^T \mathbb{E}\left[M^2(1+|X_s|)^2\right]ds.\end{aligned}$$

Therefore, for some constants C_3, C_4 and C_5 we conclude that:

$$\begin{aligned}\mathbb{E}\left[\sup_{0\le t\le T}|X_t|^2\right] &\le C_3\mathbb{E}\left[|\eta|^2\right]+C_4+C_5\int_0^T E_s ds\\ &\le \left(C_3\mathbb{E}\left[|\eta|^2\right]+C_4\right)e^{C_5T}, \quad \text{by Grönwall's Inequality.}\end{aligned}$$

We obtain the required form of (2.15), which concludes the proof in the case $p=2$.

□

Proposition 2.2.7 (Exponential Moments)**.** *For all x,t and $|\eta|\le k$, if $|\sigma(x,t)|\le k$ then:*

$$\mathbb{E}\left[\exp\left(\alpha\sup_{0\le t\le T}|X_t|\right)\right]<\infty, \quad \forall\alpha \tag{2.16}$$

and,

$$\mathbb{P}\left(\sup_{0\le t\le T}|X_t|>R\right)\le e^{-CR^2}, \quad \textit{for some constant } C \textit{ and } R \textit{ large.} \tag{2.17}$$

Proposition 2.2.8 (Hölder Continuity)**.** *Under Assumptions (A) and $\eta\in L^p$ for all p:*

$$\mathbb{E}\left[|X_t-X_s|^p\right]\le C|t-s|^{p/2}\Longrightarrow \|X_t-X_s\|_{L^p}\le C|t-s|^{1/2}. \tag{2.18}$$

By Kolmogorov's Theorem, X_t is a.s. $C^{0,\gamma}$ for $\gamma<1/2$.

Proposition 2.2.9 (Continuity w.r.t. Initial Condition)**.** *Under Assumptions (A) let X_t,Y_t be solutions of SDEs with initial conditions $X_0=\eta$ and $Y_0=\nu$. Then:*

$$\mathbb{E}\left[\sup_{0\le t\le T}|X_t-Y_t|^p\right]\le C\mathbb{E}[|\eta-\nu|^p]. \tag{2.19}$$

2.3 Generator of an SDE and Semigroups

In this section, we write $\{\xi_s^{x,t}\}$ to be the stochastic process starting in position x at time t. We also write C_b^o to be the set of continuous and bounded functions.

We now consider the following SDE:

$$\begin{cases} d\xi_s^{x,t} = b(\xi_s^{x,t}, s)ds + \sigma(\xi_s^{x,t}, s)dB_s, \quad s \geq t \\ \xi_t^{x,t} = x \end{cases} \tag{2.20}$$

Consider now the family of operators $T_{t,s} : C_b^o(\mathbb{R}^d) \longrightarrow C_b^o(\mathbb{R}^d)$ where for $u \in C_b^o(\mathbb{R}^d)$ we define:

$$(T_{t,s}u)(x) := \mathbb{E}\left[u(\xi_s^{x,t})\right]. \tag{2.21}$$

Definition 2.3.1. *A family of bounded linear operators $T(t) : E \to E$, where E is Banach space, is said to be a **semigroup** if $T(0) = \mathbb{1}$ and $T(t)T(s)f = T(t+s)f$, for all $f \in E$ and $t, s \geq 0$.*

Proposition 2.3.2. *$\{T_{t,s}\}_{t,s}$ is a semigroup.*

Proof. By the properties of a semigroup we need to verify that, for all $t \leq r \leq s$:

$$T_{t,s} = T_{t,r} \circ T_{r,s}. \tag{2.22}$$

Now using the Tower Property we get:

$$\mathbb{E}\left[u(\xi_s^{x,t})\right] = \mathbb{E}\left[\mathbb{E}\left[u\left(\xi_s^{x,t}\right)|\mathcal{F}_r\right]\right] = \mathbb{E}\left[\mathbb{E}\left[u\left(\xi_s^{\xi_r^{x,t},r}\right)\right]\right] = \mathbb{E}\left[(T_{t,r}u)(\xi_r^{x,t})\right]. \tag{2.23}$$

Therefore it holds that $(T_{t,s}u)(x) = (T_{t,r} \circ T_{r,s}u)(x)$, which implies that $\{T_{t,s}\}_{t,s}$ is a semigroup. □

Definition 2.3.3. *For $\mathcal{L}_t$: function space $\longrightarrow$ function space, we define the **generator of a semigroup** as:*

$$(\mathcal{L}_t u)(x) := \lim_{h \to 0} \frac{(T_{t,t+h}u)(x) - u(x)}{h} = \lim_{s \to t^+} \frac{1}{s-t}\left(\mathbb{E}\left[u(\xi_s^{x,t})\right] - u(x)\right). \tag{2.24}$$

Proposition 2.3.4. *For a new process $Y_s := u(\xi_s^{x,t})$ we have that:*

$$(\mathcal{L}_t u)(x) = b(x,t) \cdot \nabla u(x) + \frac{1}{2}\sigma(x,t)\sigma^T(x,t) : \nabla^2 u(x), \tag{2.25}$$

where $\mathcal{L}_t : C^2 \longrightarrow$ measurable function.

Proof. We will apply Itô Formula to the process $Y_s := u(\xi_s^{x,t})$.

$$\begin{aligned} dY_s &= \sum_i \frac{\partial u}{\partial x_i} d\left(\xi_s^{x,t}\right)^{(i)} + \frac{1}{2}\sum_{i,j} \frac{\partial^2 u}{\partial x_i \partial x_j} d\left\langle \left(\xi_s^{x,t}\right)^{(i)}, \left(\xi_s^{x,t}\right)^{(j)} \right\rangle \\ &= b(\xi_s^{x,t}, s)\nabla u(\xi_s^{x,t})ds + \frac{1}{2}\sigma(\xi_s^{x,t}, s)\sigma^T(\xi_s^{x,t}, s) : \nabla^2 u(\xi_s^{x,t}, s)ds + \nabla^T u(\xi_s^{x,t})\sigma(\xi_s^{x,t}, s)dB_s. \end{aligned}$$

Then we have:

$$Y_s = Y_t + \int_t^s \left(b\nabla u + \frac{1}{2}\sigma\sigma^2 : \nabla^2 u \right) d\tau + \int_s^t \nabla^T u \sigma dB_\tau. \tag{2.26}$$

Therefore, by the definition of the generator:

$$\begin{aligned}(\mathcal{L}_t u)(x) :&= \lim_{s\to t^+} \frac{1}{s-t}\left(\mathbb{E}\left[Y_s\right] - Y_t\right) \\ &= \lim_{s\to t^+} \frac{1}{s-t}\mathbb{E}\left[\int_t^s \left(b\nabla u + \frac{1}{2}\sigma\sigma^2 : \nabla^2 u \right) d\tau\right] \\ &= b(x,t)\nabla u(x) + \frac{1}{2}\sigma(x,t)\sigma^T(x,t) : \nabla^2 u(x),\end{aligned}$$

where we have used the Mean-Value Theorem in the last equality.

□

2.4 Feynman-Kac Formulas

In this section, we present different forms of the Feynman-Kac Formula. We will first present it in its most basic form, before presenting a more evolved version. We will finally end on a formula which has a dependence on a stopping time. We will denote by $C^{a,b}$ the set of functions $f(x,t)$ that are a times differentiable in x and b times differentiable in t.

2.4.1 Feynman-Kac Formula

In this subsection, we consider $\phi : \mathbb{R}^d \longrightarrow \mathbb{R}$ and $f : \mathbb{R}^d \times \mathbb{R}_+ \longrightarrow \mathbb{R}$ which have:

- Polynomial growth: $|\phi(x)| \leq M(1+|x|^\gamma)$ and $|f(x,t)| \leq M(1+|x|^\gamma)$, for some $\gamma > 0$.
- Exponential growth in the case σ is bounded.

Theorem 2.4.1 (Feynman-Kac Formula (1))**.** *Let $u = u(x,t) \in C^{2,1}(\mathbb{R}^d \times [0,T])$ be a solution of the PDE:*

$$\begin{cases} \partial_t u + \mathcal{L}_t u = f \quad on\ \mathbb{R}^d \times [0,T) \\ u(x,T) = \phi(x) \end{cases} \tag{2.27}$$

Then the solution to the above PDE is given by:

$$u(x,t) = \mathbb{E}\left[\phi\left(X_T^{x,t}\right)\right] - \mathbb{E}\left[\int_t^T f(X_s^{x,t}, s)ds\right]. \tag{2.28}$$

Sketch of Proof. Consider the process $Y_s = u(X_s^{x,t}, s)$ for $t \leq s \leq T$. For convenience we set $A(x,t) = \sigma(x,t)\sigma^T(x,t)$. We now use Itô Formula:

$$\begin{aligned} dY_s &= \partial_t u(X_s^{x,t}, s)ds + \nabla u(X_s^{x,t}, s)\cdot dX_s^{x,t} + \frac{1}{2}\nabla^2 u(X_s^{x,t}, s) : d\langle X_s^{x,t}\rangle_s \\ &= \partial_t u(X_s^{x,t}, s)ds + b(X_s^{x,t}, s)\cdot\nabla u(X_s^{x,t}, s)ds + \underbrace{\nabla^T u(X_s^{x,t}, s)\sigma(X_s^{x,t}, s)dB_s}_{\text{Square Integrable}} + \frac{1}{2}A(X_s^{x,t}, s) : \nabla^2 u(X_s^{x,t}, s)ds. \end{aligned}$$

Therefore:

$$\mathbb{E}\left[\phi\left(X_T^{x,t}\right)\right] = \mathbb{E}[Y_T] = \mathbb{E}[Y_t] + \mathbb{E}\left[\int_t^T \left(\partial_t u(X_s^{x,t},s) + \mathcal{L}_s u(X_s^{x,t},s)\right) ds\right]$$
$$= u(x,t) + \mathbb{E}\left[\int_t^T f(X_s^{x,t},s) ds\right].$$

□

Theorem 2.4.2. *Under Assumptions (A) or (B), if:*
- $A(x,t)$ *is uniformly elliptic (positive definite matrix), i.e.* $\exists \lambda > 0 : v^T A(x,t)\sigma \geq \lambda |v|^2, \forall x \in \mathbb{R}^d, \forall t \in [0,T], \forall v \in \mathbb{R}^d$.
- ϕ *is continuous with polynomial growth (or exponential growth if* σ *is bounded).*
- f *is locally Hölder continuous in* x *and with polynomial (or exponential) growth.*

Then, there exists a unique $u \in C^{2,1}(\mathbb{R}^d \times [0,T])$ *solution of (*2.27*) and (*2.28*).*

2.4.2 More Complex Feynman-Kac Formula

We consider the same setup as the previous subsection.

Theorem 2.4.3 (Feynman-Kac Formula (2))**.** *Let* $c : \mathbb{R}^d \times \mathbb{R}_+ \longrightarrow \mathbb{R}$ *locally Hölder and bounded from below. Consider the PDE:*

$$\begin{cases} \partial_t u + \mathcal{L}_t u - cu = f \quad on \ \mathbb{R}^d \times [0,T) \\ u(x,T) = \phi(x) \end{cases} \tag{2.29}$$

Then there exists a unique $u \in C^{2,1}(\mathbb{R}^d \times [0,T])$ *such that:*

$$u(x,t) = \mathbb{E}\left[\phi\left(X_T^{x,t}\right) Z_T\right] - \mathbb{E}\left[\int_t^T f(X_s^{x,t},s) Z_s ds\right], \tag{2.30}$$

where we have the integrating factor $Z_T = \exp\{-\int_t^T c(X_s^{x,t},s) ds\}$.

Sketch of Proof. One can prove the above characterization by considering the process $Y_s = u(X_s^{x,t},s) Z_s$ and apply Itô Formula.

□

2.4.3 Feynman-Kac Formula with Stopping Time

Theorem 2.4.4 (Feynman-Kac Formula with Stopping Time)**.** *Let $D \subset \mathbb{R}^d$ open and bounded domain, with a C^2 boundary (or Lipschitz, convex domain). Consider the PDE:*

$$\begin{cases} \partial_t u + \mathcal{L}_t u - cu = f \quad on \;\; D \times [0,T) \\ u(x,T) = \phi(x) \\ u(x,t) = g(x,t) \quad for \;\; x \in \partial D \end{cases} \tag{2.31}$$

Suppose ϕ is continuous on $\overline{D}$, g continuous function on $\partial D \times [0,T]$ and $g(x,T) = \phi(x)$. Then there exists a unique $u \in C^{2,1}(D \times [0,T]) \cap C^o(\overline{D} \times [0,T])$ such that:

$$u(x,t) = \mathbb{E}\left[\widetilde{\phi}\left(X^{x,t}_{\tau \wedge T}, \tau \wedge T\right) Z_{\tau \wedge T}\right] - \mathbb{E}\left[\int_t^{\tau \wedge T} f(X^{x,t}_s, s) Z_s ds\right], \tag{2.32}$$

where we have a stopping time $\tau = \inf\{s : X^{x,t}_s \notin D\}$ and:

$$\widetilde{\phi}(x,t) = \begin{cases} g(x,t), & t \leq T \\ \phi(x), & t = T \end{cases} . \tag{2.33}$$

2.5 Fokker-Planck Equation

In this section, we write C^∞_o for the set of C^∞ functions with compact support. Consider the SDE:

$$\begin{cases} dX_t = b(X_t,t)dt + \sigma(X_t,t)dB_t \\ X_0 = \eta \end{cases} \tag{2.34}$$

Now, let $\mu_t = \text{Law}(X_t)$ and suppose that μ_t is absolutely continuous with respect to the Lebesgue measure and denote by $p(x,t)$ the density of μ_t. Can we write an equation for $\partial_t p$? The answer is yes!

Theorem 2.5.1 (Fokker-Planck Equation)**.** *For all $x \in \mathbb{R}^d$, all $t \in [0,T]$ and $A = \sigma\sigma^T$, the equation is given by:*

$$\partial_t p(x,t) = \mathcal{L}^*_t p(x,t), \tag{2.35}$$

where we have that $\mathcal{L}^*_t p := -\nabla \cdot b(x,t)p(x,t) + \frac{1}{2}\nabla^2 : (A(x,t)p(x,t))$.

Sketch of Proof. Take $\phi \in C^\infty_o(\mathbb{R}^d \times [0,T])$ and consider the process $Y_t = \phi(X_t,t)$. Using Itô Formula we have:

$$Y_T = Y_0 + \int_0^T \left(\partial_t \phi(X_s,s) + \mathcal{L}_t \phi(X_s,s)\right) ds + \int_0^T \underbrace{\nabla \phi A}_{\text{Square Integrable}} \quad dB_s. \tag{2.36}$$

Hence:

$$\underbrace{\mathbb{E}(Y_T)}_{=0} = \underbrace{\mathbb{E}(Y_0)}_{=0} + \mathbb{E}\left[\int_0^T (\partial_t + \mathcal{L}_t)\phi(X_s, s)ds\right], \quad \forall \phi \in C_o^\infty. \tag{2.37}$$

We now use Fubini's Theorem to interchange expectation and integral:

$$\begin{aligned}\mathbb{E}\left[\int_0^T (\partial_t + \mathcal{L}_t)\phi(X_s, s)ds\right] &= \int_0^T \mathbb{E}\left[(\partial_t + \mathcal{L}_t)\phi(X_s, s)\right] ds \\ &= \int_0^T \int_{\mathbb{R}^d} (\partial_t + \mathcal{L}_t)\phi(x, s)p(x, s)dxds = 0.\end{aligned}$$

Now integrating by parts we obtain:

$$\int_0^T \int_{\mathbb{R}^d} (\partial_t + \mathcal{L}_t)\phi(x, s)p(x, s)dxds = \int_0^T \int_{\mathbb{R}^d} \phi(x, t)(-\partial_t + \mathcal{L}_t^*)p(x, t)dxdt = 0. \tag{2.38}$$

For the second integral to vanish, we require $\partial_t p(x,t) = \mathcal{L}_t^* p(x,t)$.

□

Chapter 3

Numerical Methods for Stochastic Differential Equations

We now begin the numerical analysis of SDEs where we first provide some motivation.
Consider the SDE:

$$\begin{cases} dX_t = b(X_t,t)dt + \sigma(X_t,t)dB_t, & 0 \le t \le T \\ X_0 = \eta \end{cases} \tag{3.1}$$

We still assume that b, σ are globally Lipschitz and Linear Growth Bound.
Furthermore, we set a time grid $0 = t_0 < t_1 < ... < t_N = T$ with $\Delta t_n = t_{n+1} - t_n$ and $\Delta t = \max_{0 \le n \le N} \Delta t_n$.
We seek for a discrete time process $\{Y_n\}_{n=0}^N$ such that $Y_n \approx X(t_n)$. Now the exact process satisfies:

$$X(t_{n+1}) = X(t_n) + \underbrace{\int_{t_n}^{t_{n+1}} b(X_s,s)ds}_{\approx b(X(t_n),t_n)\Delta t_n} + \underbrace{\int_{t_n}^{t_{n+1}} \sigma(X_s,s)dB_s}_{\approx \sigma(X(t_n),t_n)\Delta W_n}, \quad \text{for } \Delta W_n = W_{t_{n+1}} - W_{t_n}. \tag{3.2}$$

- If we use an Euler-Maruyama Scheme we obtain:

$$Y_{n+1} = Y_n + b(Y_n,t_n)\Delta t_n + \sigma(Y_n,t_n)\Delta W_n. \tag{3.3}$$

- If we use a Stochastic θ-Scheme we obtain:

$$Y_{n+1} = Y_n + (\theta b(Y_{n+1},t_{n+1}) + (1-\theta)b(Y_n,t_n))\,\Delta t_n + \underbrace{\sigma(Y_n,t_n)\Delta W_n}_{(*)}. \tag{3.4}$$

Note that we cannot replace $(*)$ with $(\theta\sigma(Y_{n+1},t_{n+1}) + (1-\theta)\sigma(Y_n,t_n))$ since we would not respect the Itô definition of integration. Furthermore, a second problem could arise. If we consider $dX_t = \sigma X_t dB_t$ for $\theta = 1$ we find $Y_{n+1} = Y_n + \sigma Y_{n+1}\Delta W_n \Longrightarrow Y_{n+1} = \frac{1}{1+\sigma\Delta W_n} Y_n$. However, if we want to compute $\mathbb{E}(Y_{n+1})$ we could have that the integrand is not integrable, especially in the case where σ is very small. We will see how to resolve that problem later on.

Definition 3.0.1. *We say that $\{Y_n\}_n$ **converges strongly** to $\{X_t\}_t$ if ($p \ge 1$):*

$$\mathbb{E}\left[\sup_{0 \le n \le N} |Y_n - X(t_n)|^p\right]^{1/p} \xrightarrow{\Delta t \to 0} 0. \tag{3.5}$$

Definition 3.0.2. *We say that* $\{Y_n\}_n$ *has* ***strong order*** **k** *if* ($p \geq 1$, $C \in \mathbb{R}$)*:*

$$\mathbb{E}\left[\sup_{0\leq n\leq N} |Y_n - X(t_n)|^p\right]^{1/p} \leq C(\Delta t)^k. \tag{3.6}$$

Definition 3.0.3. *We say that* $\{Y_n\}_n$ ***converges weakly*** *to* $\{X_t\}_t$ *if, for* ϕ *sufficiently smooth:*

$$|\mathbb{E}[\phi(Y_N) - \phi(X(T))]| \xrightarrow{\Delta t\to 0} 0. \tag{3.7}$$

Definition 3.0.4. *We say that* $\{Y_n\}_n$ *has* ***weak order*** **k** *if* ($C \in \mathbb{R}$)*:*

$$|\mathbb{E}[\phi(Y_N) - \phi(X(T))]| \leq C(\Delta t)^k. \tag{3.8}$$

3.1 The Euler-Maruyama Scheme

We consider, as usual, a d-dimensional SDE:

$$\begin{cases} dX_t = b(X_t,t)dt + \sigma(X_t,t)dW_t, & t \in (0,T] \\ X_0 = \eta \end{cases} \tag{3.9}$$

where $\{W_t\}_t$ is a m-dimensional Brownian Motion, $b : \mathbb{R}^d \times \mathbb{R}_+ \longrightarrow \mathbb{R}^d$ and $\sigma : \mathbb{R}^d \times \mathbb{R}_+ \longrightarrow \mathbb{R}^{d\times m}$.
We furthermore consider the usual assumptions:
a) Global Lipschitz Condition: There exists $L > 0$ such that:

$$|b(x,t) - b(y,t)| + \|\sigma(x,t) - \sigma(y,t)\|_F \leq L|x-y|, \quad \forall t. \tag{3.10}$$

b) Linear Growth Bound: There exists $M > 0$ such that:

$$|b(x,t)| + \|\sigma(x,t)\|_F \leq M(1+|x|). \tag{3.11}$$

c) $\eta \in L^2(\Omega)$ (or $L^p(\Omega)$ for $p \geq 2$).

d) b, σ depend smoothly on t, at least $\frac{1}{2}$-Hölder, where there exists $\kappa > 0$ such that:

$$|b(x,t) - b(x,s)| + \|\sigma(x,t) - \sigma(x,s)\|_F \leq \kappa(1+|x|)|t-s|^{\frac{1}{2}}. \tag{3.12}$$

Definition 3.1.1 (The Euler-Maruyama Scheme)**.** *Given a partition* $0 = t_0 < t_1 < ... < t_N = T$*, writting* $\Delta t_n = t_{n+1} - t_n$ *and* $\Delta W_n = W_{n+1} - W_n$*, the scheme is given by:*

$$\begin{cases} Y_{n+1} = Y_n + b(Y_n,t_n)\Delta t_n + \sigma(Y_n,t_n)\Delta W_n \\ Y_0 = \eta \end{cases}. \tag{3.13}$$

Now recall the L^p bound in Proposition (2.2.6) along the Hölder continuity condition for $p = 2$ in Proposition (2.2.8). We now present similar bounds for discrete processes.

Lemma 3.1.2. *Under conditions **b)** and **c)**, if $\mathbb{E}[|\eta|^p] < \infty$ for all $p \geq 2$ then for all n, the discrete process $\{Y_n\}_{n=0}^N$ satisfies:*

$$\mathbb{E}[|Y_n|^p] \leq \widehat{C}(T,M)(1+\mathbb{E}[|\eta|^p]). \tag{3.14}$$

Proof. For $p=2$. We use induction. Assume that $\mathbb{E}[|Y_n|^2] \leq \widehat{C}(T,M)(1+\mathbb{E}[|\eta|^2])$, which is true for $n=0$ and we specify $\widehat{C}(T,M)$ later on.
For simplicity, let $b_n := b(Y_n,t_n)$ and $\sigma_n := \sigma(Y_n,t_n)$. Now:

$$\begin{aligned}\mathbb{E}\left[|Y_{n+1}|^2\right] &= \mathbb{E}\left[|Y_n + b_n\Delta t_n + \sigma_n \Delta W_n|^2\right] \\ &= \mathbb{E}\left[|Y_n|^2\right] + \underbrace{\mathbb{E}\left[|b_n|^2\right](\Delta t_n)^2}_{A} + \underbrace{\mathbb{E}\left[|\sigma_n\Delta W_n|^2\right]}_{B} + \underbrace{2\mathbb{E}\left[Y_n^T b_n \Delta t_n\right]}_{C} + 2\mathbb{E}\left[Y_n^T\sigma_n\Delta W_n\right] \\ &\qquad + 2\mathbb{E}\left[b_n^T\sigma_n\Delta W_n\Delta t_n\right].\end{aligned}$$

Now note that $Y_n^T\sigma_n$ and $b_n^T\sigma_n$ are integrable, and that $\mathbb{E}\left[Y_n^T\sigma_n\mathbb{E}[\Delta W_n|\mathcal{F}_n]\right] = 0$. We then find that:

$$\mathbb{E}\left[Y_n^T\sigma_n\Delta W_n\right] = 0 \quad \text{and,} \quad \mathbb{E}\left[b_n^T\sigma_n\Delta W_n\Delta t_n\right] = 0. \tag{3.15}$$

We now focus on the terms A, B and C.

- Starting with term A we find:

$$A = \mathbb{E}\left[|b_n|^2\right](\Delta t_n)^2 \leq M^2\mathbb{E}\left[(1+|Y_n|)^2\right](\Delta t_n)^2 \leq 2M^2\mathbb{E}\left[1+|Y_n|^2\right]\Delta t_n T. \tag{3.16}$$

- Now focusing on term B we have:

$$B = \mathbb{E}\left[|\sigma_n\Delta W_n|^2\right] = \mathbb{E}\left[\sum_{i=1}^d\left(\sum_{k=1}^m \sigma_n^{(i,k)}\Delta W_n^{(k)}\right)^2\right] = \sum_{i=1}^d\sum_{k,k'=1}^m \mathbb{E}\left[\sigma_n^{(i,k)}\sigma_n^{(i,k')}\Delta W_n^{(k)}\Delta W_n^{(k')}\right]. \tag{3.17}$$

Remark that $\mathbb{E}\left[\Delta W_n^{(k)}\Delta W_n^{(k')}\right] = \delta_{kk'}\Delta t_n$. Hence B becomes:

$$B = \sum_{i=1}^d\sum_{k=1}^m \mathbb{E}\left[\left(\sigma_n^{(i,k)}\right)^2\right]\Delta t_n = \mathbb{E}\left[\|\sigma_n\|_F^2\right]\Delta t_n \leq M^2\mathbb{E}\left[(1+|Y_n|)^2\right]\Delta t_n \leq 2M^2\mathbb{E}\left[1+|Y_n|^2\right]\Delta t_n. \tag{3.18}$$

- Finally for term C we obtain:

$$C = 2\mathbb{E}\left[Y_n^T b_n\Delta t_n\right] \leq 2\mathbb{E}\left[|Y_n||b_n|\Delta t_n\right] \leq 2M\mathbb{E}\underbrace{\left[|Y_n|(1+|Y_n|)\right]}_{\leq(1+|Y_n|)^2}\Delta t_n \leq 4M\Delta t_n\mathbb{E}\left[1+|Y_n|^2\right]. \tag{3.19}$$

Therefore, regrouping everything together, we find:

$$\begin{aligned}\mathbb{E}\left[1+|Y_{n+1}|^2\right] &\leq \mathbb{E}\left[1+|Y_n|^2\right] + \Delta t_n\mathbb{E}\left[1+|Y_n|^2\right]\underbrace{(2M^2T+M^2+4M)}_{=\widetilde{C}} \\ &= \left(1+\widetilde{C}\Delta t_n\right)\mathbb{E}\left[1+|Y_n|^2\right]\end{aligned}$$

$$= \prod_{j=1}^{n} \underbrace{\left(1+\widetilde{C}\Delta t_j\right)}_{\exp(\widetilde{C}\Delta t_i)} \mathbb{E}\left[1+|\eta|^2\right] \leq e^{\widetilde{C}T}\mathbb{E}\left[1+|\eta|^2\right].$$

This leads us to the required result:

$$\mathbb{E}\left[|Y_{n+1}|^2\right] \leq e^{\widetilde{C}T}\mathbb{E}\left[1+|\eta|^2\right] \equiv \widehat{C}(T,M)\mathbb{E}\left[1+|\eta|^2\right]. \tag{3.20}$$

□

Definition 3.1.3 (The Interpolated Euler-Maruyama Scheme)**.** *For all $t_n < t < t_{n+1}$, the interpolated scheme is given by:*

$$\widehat{Y}_t = Y_n + b(Y_n, t_n)(t-t_n) + \sigma(Y_n, t_n)(W_t - W_{t_n}). \tag{3.21}$$

Theorem 3.1.4. *Under Assumptions **a)-d)** and denoting by $\Delta t = \max_n(\Delta t_n)$, there exists $C > 0$ such that:*

$$\mathbb{E}\left[\sup_{0\leq n\leq N}|X_{t_n} - Y_n|^2\right] \leq \mathbb{E}\left[\sup_{0\leq t\leq T}|X_t - \widehat{Y}_t|^2\right] \leq C\Delta t. \tag{3.22}$$

Before we present the proof, we introduce some notation and setup. First define $n_s = \max\{n : t_n < s\}$ and let $\phi_s = t_{n_s}$. Then it holds that $Y_{n_s} = \widehat{Y}_{\phi_s}$. We can now rewrite $\widehat{Y}_t$ as:

$$\widehat{Y}_t = \eta + \int_0^t b(Y_{n_s}, \phi_s)ds + \int_0^t \sigma(Y_{n_s}, \phi_s)dW_s, \tag{3.23}$$

with original SDE:

$$X_t = \eta + \int_0^t b(X_s, s)ds + \int_0^t \sigma(X_s, s)dW_s. \tag{3.24}$$

Taking a difference of the two above integrals yields:

$$X_t - \widehat{Y}_t = \int_0^t (b(X_s,s) - b(Y_{n_s},\phi_s))ds + \int_0^t (\sigma(X_s,s) - \sigma(Y_{n_s},\phi_s))dW_s. \tag{3.25}$$

We consider an intermediate solution, for $t_n < t < t_{n+1}$,

$$\widehat{X}_t = X_{t_n} + b(X_{t_n}, t_n)(t-t_n) + \sigma(X_{t_n}, t_n)(W_t - W_{t_n}). \tag{3.26}$$

Now, we separate (3.25) with $\pm b(X_{\phi_s}, \phi_s)$ and $\pm\sigma(X_{\phi_s}, \phi_s)$ in the following way:

$$X_t - \widehat{Y}_t = \underbrace{\int_0^t (b(X_{\phi_s},\phi_s) - b(Y_{n_s},\phi_s))ds + \int_0^t (\sigma(X_{\phi_s},\phi_s) - \sigma(Y_{n_s},\phi_s))dW_s}_{T_t^{(1)}}$$
$$+ \underbrace{\int_0^t (b(X_s,s) - b(X_{\phi_s},\phi_s))ds + \int_0^t (\sigma(X_s,s) - \sigma(X_{\phi_s},\phi_s))dW_s}_{T_t^{(2)}}.$$

We note that $T_t^{(2)}$ is called the cumulated local truncation error.

Proof. The idea of the proof is to break the expectation in parts that we will evaluate and bound independently. We set:

$$e_T := \mathbb{E}\left[\sup_{0\le t\le T}|X_t-\widehat{Y}_t|^2\right] = \mathbb{E}\left[\sup_{0\le t\le T}\left|T_t^{(1)}+T_t^{(2)}\right|^2\right]$$
$$\le \underbrace{2\mathbb{E}\left[\sup_{0\le t\le T}\left|T_t^{(1)}\right|^2\right]}_{A} + \underbrace{2\mathbb{E}\left[\sup_{0\le t\le T}\left|T_t^{(2)}\right|^2\right]}_{B}.$$

Upper-bound for A: We again decompose the integral into two parts that we will bound independently.

$$A = 2\mathbb{E}\left[\sup_{0\le t\le T}\left|T_t^{(1)}\right|^2\right] \le \underbrace{4\mathbb{E}\left[\sup_{0\le t\le T}\left|\int_0^t (b(X_{\phi_s},\phi_s)-b(Y_{n_s},\phi_s))ds\right|^2\right]}_{T_t^{(1,1)}}$$
$$+\underbrace{4\mathbb{E}\left[\sup_{0\le t\le T}\left|\int_0^t (\sigma(X_{\phi_s},\phi_s)-\sigma(Y_{n_s},\phi_s))ds\right|^2\right]}_{T_t^{(1,2)}}.$$

- We first start with $T_t^{(1,1)}$:

$$\begin{aligned}
T_t^{(1,1)} &\le 4\mathbb{E}\left[\sup_{0\le t\le T} t\int_0^t |(b(X_{\phi_s},\phi_s)-b(Y_{n_s},\phi_s))|^2\,ds\right], \quad \text{by Cauchy-Schwarz,}\\
&\le 4\mathbb{E}\left[T\int_0^T |(b(X_{\phi_s},\phi_s)-b(Y_{n_s},\phi_s))|^2\,ds\right]\\
&\le 4TL^2\int_0^T \mathbb{E}\left[|X_{\phi_s}-Y_{n_s}|^2\right]ds\\
&\le 4TL^2\int_0^T \mathbb{E}\left[\sup_{0\le t\le T}|X_t-\widehat{Y}_t|^2\right]ds = 4TL^2\int_0^T e_s ds.
\end{aligned}$$

- We continue with $T_t^{(1,2)}$:

$$\begin{aligned}
T_t^{(1,2)} &\le 4\mathbb{E}\left[\sup_{0\le t\le T}\left|\int_0^t (\sigma(X_{\phi_s},\phi_s)-\sigma(Y_{n_s},\phi_s))dW_s\right|^2\right]\\
&\le 16\mathbb{E}\left[\left|\int_0^T (\sigma(X_{\phi_s},\phi_s)-\sigma(Y_{n_s},\phi_s))dW_s\right|^2\right], \quad \text{by Doob's Inequality,}\\
&= 16\mathbb{E}\left[\int_0^T \|(\sigma(X_{\phi_s},\phi_s)-\sigma(Y_{n_s},\phi_s))\|_F^2\,ds\right]\\
&\le 16L^2\mathbb{E}\left[\int_0^T |X_{\phi_s}-Y_{n_s}|^2 ds\right]\\
&= 16L^2\int_0^T \mathbb{E}\left[|X_{\phi_s}-Y_{n_s}|^2\right]ds\\
&\le 16L^2\int_0^T \mathbb{E}\left[\sup_{0\le t\le T}|X_t-\widehat{Y}_t|^2\right]ds = 16L^2\int_0^T e_s ds.
\end{aligned}$$

Upper-bound for B: Similarly to the previous part we once more decompose B into two parts that we will bound independently.

$$B = 2\mathbb{E}\left[\sup_{0\le t\le T}\left|T_t^{(2)}\right|^2\right] \le \underbrace{4\mathbb{E}\left[\sup_{0\le t\le T}\left|\int_0^t (b(X_s,s)-b(X_{\phi_s},\phi_s))ds\right|^2\right]}_{T^{(2,1)}} + \underbrace{4\mathbb{E}\left[\sup_{0\le t\le T}\left|\int_0^t (\sigma(X_s,s)-\sigma(X_{\phi_s},\phi_s))ds\right|^2\right]}_{T^{(2,2)}}.$$

- We start with $T^{(2,1)}$ by including $\pm b(X_s,\phi_s)$:

$$\begin{aligned} T^{(2,1)} &\le 4\mathbb{E}\left[T\int_0^T |(b(X_s,s)-b(X_{\phi_s},\phi_s))|^2 ds\right] \\ &\le 8\mathbb{E}\left[T\int_0^T \underbrace{|(b(X_s,s)-b(X_s,\phi_s))|^2}_{\le \kappa^2(1+|X_s|)^2|s-\phi_s|} ds + T\int_0^T \underbrace{|(b(X_s,\phi_s)-b(X_{\phi_s},\phi_s))|^2}_{\le L^2|X_s-X_{\phi_s}|^2} ds\right]. \end{aligned}$$

We now note that $|s-\phi_s| = \Delta t$ and $\mathbb{E}\left[|X_s - X_{\phi_s}|^2\right] \le C|s-\phi_s| = \mathcal{O}(\Delta t)$. Hence, $T^{(2,1)} \le C\Delta t$.

- Finally we end with $T^{(2,2)}$ by including $\pm\sigma(X_s,\phi_s)$:

$$\begin{aligned} T^{(2,2)} &\le 16\mathbb{E}\left[\left|\int_0^T (\sigma(X_s,s)-\sigma(X_{\phi_s},\phi_s))dW_s\right|^2\right], \quad \text{by Doob's Inequality,} \\ &= 16\mathbb{E}\left[\int_0^T \|\sigma(X_s,s)-\sigma(X_{\phi_s},\phi_s)\|_F^2 ds\right], \quad \text{by Itô Isometry,} \\ &\le C\Delta t. \end{aligned}$$

Putting everything together and using Grönwall's Lemma yields:

$$e_T \le 4L^2(4+T)\int_0^T e_s ds + C\Delta t \le e^{4L^2(4+T)}C\Delta t, \quad \text{as required.} \tag{3.27}$$

□

Using the Burkholder-Davis-Gundy Inequality, one can show a generalization of the previous Theorem.

Theorem 3.1.5. *Consider $\eta \in L^p$ and $p \ge 2$. Under Assumptions **a)-d)** and denoting by $\Delta t = \max_n(\Delta t_n)$, there exists $C > 0$ such that:*

$$\mathbb{E}\left[\sup_{0\le t\le T}|X_t - \widehat{Y}_t|^p\right]^{1/p} \le C(\Delta t)^{\frac{1}{2}}. \tag{3.28}$$

Furthermore, we can present an even more powerful theorem that gets rid of the expectation value from the previous theorems, and gives us an almost sure statement of strong convergence.

Theorem 3.1.6. • *If $\eta \in L^p$ for $p \geq 2$:*

$$N^\alpha \sup_{0\leq t\leq T} |X_t - \widehat{Y}_t| \xrightarrow{N\to\infty} 0 \quad a.s. \text{ if } \quad \alpha < \frac{1}{2} - \frac{1}{p}. \tag{3.29}$$

• *If $\eta \in L^p$ for <u>all</u> $p \geq 2$:*

$$N^\alpha \sup_{0\leq t\leq T} |X_t - \widehat{Y}_t| \xrightarrow{N\to\infty} 0 \quad a.s. \text{ for } \underline{all} \quad \alpha < \frac{1}{2}. \tag{3.30}$$

Proof. Set $Z_N = N^\alpha \sup_{0\leq t\leq T}|X_t - \widehat{Y}_t^N|$. Then by Markov's Inequality we have:

$$\mathbb{P}(Z_N > \epsilon) \leq \frac{\mathbb{E}\left[|Z_N|^p\right]}{\epsilon^p} = \frac{N^{p\alpha}\mathbb{E}\left[\sup_{0\leq t\leq T}|X_t - \widehat{Y}_t^N|\right]}{\epsilon^p} \leq \frac{C(p)}{\epsilon^p} N^{\alpha p}(\Delta t)^{\frac{p}{2}} = \frac{T^{\frac{p}{2}}C(p)}{\epsilon^p} N^{\alpha p - \frac{p}{2}}. \tag{3.31}$$

Notice that $\alpha p - \frac{p}{2} < \left(\frac{1}{2} - \frac{1}{p}\right)p - \frac{p}{2} < -1$, hence the series $\sum_{N=1}^\infty N^{\alpha p - \frac{p}{2}}$ converges. By Borel-Cantelli Lemma, the event $Z_N > \epsilon$ happens infinitely often. Since ϵ is arbitrary, we conclude that $Z_N \to 0$ almost surely. This shows the first result, and the second is an immediate consequence of the first one.

□

3.2 Error Analysis of Euler-Maruyama and the Milstein Scheme

Consider a d-dimensional SDE with m-dimensional noise:

$$\begin{cases} d\overrightarrow{X_t} = \overrightarrow{b}(\overrightarrow{X_t}, t)dt + \sigma(\overrightarrow{X_t}, t)d\overrightarrow{W_t} \\ X_0 = \eta \end{cases} \tag{3.32}$$

We again consider the Euler-Maruyama Scheme, for $t_n < t < t_{n+1}$,

$$\begin{cases} Y_{n+1} = Y_n + b(Y_n, t_n)\Delta t_n + \sigma(Y_n, t_n)\Delta W_n \\ Y_0 = \eta \end{cases}. \tag{3.33}$$

We also recall its interpolated version,

$$\begin{aligned} \widehat{Y}_t &= Y_n + b(Y_n, t_n)(t - t_n) + \sigma(Y_n, t_n)(W_t - W_{t_n}) \\ &= Y_0 + \int_0^t b(Y_{n_s}, \phi_s)ds + \int_0^t \sigma(Y_{n_s}, \phi_s)dW_s, \end{aligned}$$

with an intermediate solution,

$$\widehat{X}_t = X_{t_n} + b(X_{t_n}, t_n)(t - t_n) + \sigma(X_{t_n}, t_n)(W_t - W_{t_n}). \tag{3.34}$$

We furthermore recall the error splitting we presented in (3.25),

$$X_t - \widehat{Y}_t = \int_0^t (b(X_s, s) - b(Y_{n_s}, \phi_s))ds + \int_0^t (\sigma(X_s, s) - \sigma(Y_{n_s}, \phi_s))dW_s$$

$$= \underbrace{\int_0^t (b(X_{\phi_s},\phi_s) - b(Y_{n_s},\phi_s))ds + \int_0^t (\sigma(X_{\phi_s},\phi_s) - \sigma(Y_{n_s},\phi_s))dW_s}_{T_t^{(1)}}$$
$$+ \underbrace{\int_0^t (b(X_s,s) - b(X_{\phi_s},\phi_s))ds + \int_0^t (\sigma(X_s,s) - \sigma(X_{\phi_s},\phi_s))dW_s}_{T_t^{(2)}}.$$

Here, $T_t^{(2)}$ is the global truncation error up to t. We can subdivide the global truncation into a sum of local truncations:

$$\begin{aligned} T_T^{(2)} &= \int_0^T (b(X_s,s) - b(X_{\phi_s},\phi_s))ds + \int_0^T (\sigma(X_s,s) - \sigma(X_{\phi_s},\phi_s))dW_s \\ &= \sum_{n=0}^{N-1} \left(\int_{t_n}^{t_{n+1}} (b(X_s,s) - b(X_{t_n},t_n))ds + \int_{t_n}^{t_{n+1}} (\sigma(X_s,s) - \sigma(X_{t_n},t_n))dW_s \right) \\ &\equiv \sum_{n=0}^{N-1} T_n^{(2)}. \end{aligned}$$

In what follows, we suppose that $b, \sigma \in C^{2,1}$ with bounded derivatives. We have:

$$b(X_s,s) - b(X_{t_n},t_n) = \int_{t_n}^s db \quad \text{and,} \quad \sigma(X_s,s) - \sigma(X_{t_n},t_n) = \int_{t_n}^s d\sigma. \tag{3.35}$$

Recall that for $f \in C^{2,1}$, Itô's Formula gives us:

$$\begin{aligned} df(X_t,t) &= \left(\frac{\partial f}{\partial t}(X_t,t) + (\mathcal{L}_t f)(X_t,t) \right) dt + \sum_{k=1}^m \overrightarrow{\sigma}^{(k)}(X_t,t) \cdot \nabla f(X_t,t) dW_t^k \\ &:= \mathcal{L}_t^0 f(X_t,t)dt + \sum_{k=1}^m \mathcal{L}_t^k f(X_t,t) dW_t^k, \end{aligned}$$

where $(\mathcal{L}_t f)(x,t) = b(x,t) \cdot \nabla f(x,t) + \frac{1}{2}\sigma(x,t)\sigma^T(x,t) : \nabla^2 f(x,t)$, $\mathcal{L}_t^0 = \frac{\partial}{\partial t} + \mathcal{L}_t$ and $\mathcal{L}_t^k = \overrightarrow{\sigma}^{(k)} \cdot \nabla$. Therefore, we can write $T_n^{(2)}$ using Itô's Formula in the following way:

$$T_n^{(2)} = \underbrace{\int_{t_n}^{t_{n+1}} \int_{t_n}^s \mathcal{L}_r^0 b(X_r,r) dr ds}_{R_n^{(1)}} + \underbrace{\sum_{k=1}^m \int_{t_n}^{t_{n+1}} \int_{t_n}^s \mathcal{L}_r^k b(X_r,r) dW_r^k ds}_{R_n^{(2)}} + \underbrace{\int_{t_n}^{t_{n+1}} \int_{t_n}^s \mathcal{L}_r^0 \sigma(X_r,r) dr d\overrightarrow{W_s}}_{R_n^{(3)}}$$
$$+ \underbrace{\sum_{k=1}^m \int_{t_n}^{t_{n+1}} \int_{t_n}^s \mathcal{L}_r^k \sigma(X_r,r) dW_r^k d\overrightarrow{W_s}}_{R_n^{(4)}}.$$

Proposition 3.2.1. *Considering the total truncation error $T_T^{(2)}$, we have:*

$$\mathbb{E}\left[\left\| T_T^{(2)} \right\|^2 \right] = \mathcal{O}(\Delta t). \tag{3.36}$$

Proof. We can first write $\mathbb{E}\left[\left\|T_T^{(2)}\right\|^2\right]$ in terms of the $R_n^{(i)}$, $i=\{1,2,3,4\}$:

$$\begin{aligned}\mathbb{E}\left[\left\|T_T^{(2)}\right\|^2\right] &= \mathbb{E}\left[\left\|\sum_{n=0}^{N-1}\left(R_n^{(1)}+R_n^{(2)}+R_n^{(3)}+R_n^{(4)}\right)\right\|^2\right] \\ &\leq C\left(\mathbb{E}\left[\left\|\sum_{n=0}^{N-1}R_n^{(1)}\right\|^2\right]+\mathbb{E}\left[\left\|\sum_{n=0}^{N-1}R_n^{(2)}\right\|^2\right]+\mathbb{E}\left[\left\|\sum_{n=0}^{N-1}R_n^{(3)}\right\|^2\right]+\mathbb{E}\left[\left\|\sum_{n=0}^{N-1}R_n^{(4)}\right\|^2\right]\right).\end{aligned}$$

We will now bound each terms above to find orders of Δt.

- We first start with $R_T^{(1)}$:

$$\begin{aligned}\mathbb{E}\left[\left\|R_T^{(1)}\right\|^2\right] = \mathbb{E}\left[\left\|\sum_{n=0}^{N-1}R_n^{(1)}\right\|^2\right] &\leq \mathbb{E}\left[\left(\sum_{n=0}^{N-1}\left\|R_n^{(1)}\right\|\right)^2\right] \\ &\leq \mathbb{E}\left[N\sum_{n=0}^{N-1}\left\|R_n^{(1)}\right\|^2\right], \quad \text{by Cauchy-Schwarz,} \\ &= \frac{T}{\Delta t}\sum_{n=0}^{N-1}\mathbb{E}\left[\left\|R_n^{(1)}\right\|^2\right],\end{aligned}$$

where we have:

$$\mathbb{E}\left[\left\|R_n^{(1)}\right\|^2\right] = \mathbb{E}\left[\left\|\int_{t_n}^{t_{n+1}}\int_{t_n}^{s}\mathcal{L}_r^0 b(X_r,r)drds\right\|^2\right] \leq \mathbb{E}\Big[\int_{t_n}^{t_{n+1}}\int_{t_n}^{s}\underbrace{\left\|\mathcal{L}_r^0 b(X_r,r)\right\|^2}_{\leq\widetilde{C}}drds\Big] = \mathcal{O}[(\Delta t)^4].$$

Therefore, $\mathbb{E}\left[\left\|R_T^{(1)}\right\|^2\right] \leq \frac{T}{\Delta t}\sum_{n=0}^{N-1}O[(\Delta t)^4] = \mathcal{O}[(\Delta t)^2]$.

- Next, we continue with $R_T^{(2)}$:

$$\mathbb{E}\left[\left\|R_T^{(2)}\right\|^2\right] \leq N\mathbb{E}\left[\sum_{n=0}^{N-1}\left\|R_n^{(2)}\right\|^2\right],$$

where we have, using Itô's Isometry:

$$\begin{aligned}\mathbb{E}\left[\left\|R_n^{(2)}\right\|^2\right] &= \mathbb{E}\left[\left\|\sum_{k=1}^{m}\int_{t_n}^{t_{n+1}}\int_{t_n}^{s}\mathcal{L}_r^k b(X_r,r)dW_r^k ds\right\|^2\right] \\ &\leq \mathbb{E}\left[\int_{t_n}^{t_{n+1}}\left\|\int_{t_n}^{s}\sum_{k=1}^{m}\mathcal{L}_r^k b(X_r,r)dW_r^k\right\|^2 ds\right] \\ &\leq \Delta t\int_{t_n}^{t_{n+1}}\mathbb{E}\left(\left\|\int_{t_n}^{s}\sum_{k=1}^{m}\mathcal{L}_r^k b(X_r,r)dW_r^k\right\|^2\right)ds \\ &= \Delta t\int_{t_n}^{t_{n+1}}\left(\int_{t_n}^{s}\mathbb{E}\left(\left\|\sum_{k=1}^{m}\mathcal{L}_r^k b(X_r,r)\right\|^2\right)dr\right)ds = \mathcal{O}[(\Delta t)^3].\end{aligned}$$

Therefore, $\mathbb{E}\left[\left\|R_T^{(2)}\right\|^2\right] \leq \mathcal{O}[(\Delta t)^2]$.

- Moving on to $R_T^{(3)}$ we find:

$$\mathbb{E}\left[\left\|R_T^{(3)}\right\|^2\right] \leq N\mathbb{E}\left[\sum_{n=0}^{N-1}\left\|R_n^{(3)}\right\|^2\right],$$

where we find, using Itô's Isometry:

$$\begin{aligned}
\mathbb{E}\left[\left\|R_n^{(3)}\right\|^2\right] &= \mathbb{E}\left[\left\|\int_{t_n}^{t_{n+1}}\int_{t_n}^{s}\mathcal{L}_r^0\sigma(X_r,r)dr\overrightarrow{dW_s}\right\|^2\right] \\
&= \int_{t_n}^{t_{n+1}}\mathbb{E}\left[\left\|\int_{t_n}^{s}\mathcal{L}_r^0\sigma(X_r,r)dr\right\|^2\right]ds \\
&\leq \int_{t_n}^{t_{n+1}}\mathbb{E}\left[\int_{t_n}^{s}\left\|\mathcal{L}_r^0\sigma(X_r,r)\right\|^2 dr\right]ds \\
&\leq \Delta t\int_{t_n}^{t_{n+1}}\int_{t_n}^{s}\underbrace{\mathbb{E}\left[\left\|\mathcal{L}_r^0\sigma(X_r,r)\right\|^2\right]}_{\leq\widetilde{C}}drds = \mathcal{O}[(\Delta t)^3].
\end{aligned}$$

Therefore, $\mathbb{E}\left[\left\|R_T^{(3)}\right\|^2\right] \leq \mathcal{O}[(\Delta t)^2]$.

- Finally, it remains to bound $R_T^{(4)}$. One can show that $\mathbb{E}\left[\left\|R_n^{(4)}\right\|^2\right] = \mathcal{O}[(\Delta t)^2]$ and therefore $\mathbb{E}\left[\left\|R_T^{(4)}\right\|^2\right] = \mathcal{O}(\Delta t)$.

□

We remark that the term leading the truncation error are the last double stochastic integrals $R_T^{(4)}$ giving an order of $\mathcal{O}(\Delta t)$. We can now rewrite $R_n^{(4)}$ using Itô's Formula in the following way:

$$\begin{aligned}
R_n^{(4)} &= \int_{t_n}^{t_{n+1}}\int_{t_n}^{s}\sum_{k=1}^{m}\mathcal{L}_r^k\sigma(X_r,r)dW_r^k\overrightarrow{dW_s} \\
&= \int_{t_n}^{t_{n+1}}\int_{t_n}^{s}\left[\sum_{k=1}^{m}\left(\mathcal{L}_{t_n}^k\sigma(X_{t_n},t_n)+\int_{t_n}^{r}\mathcal{L}_u^0\left(\mathcal{L}_u^k\sigma(X_u,u)\right)du+\int_{t_n}^{r}\sum_{k'=1}^{m}\mathcal{L}_u^{k'}\left(\mathcal{L}_u^k\sigma(X_u,u)\right)dW_u^{k'}\right]dW_r^k dW_s \\
&= \sum_{k=1}^{m}\mathcal{L}_{t_n}^k\sigma(X_{t_n},t_n)\int_{t_n}^{t_{n+1}}\int_{t_n}^{s}dW_r^k\overrightarrow{dW_s}+\widetilde{R_n}^{4,1}+\widetilde{R_n}^{4,2},
\end{aligned}$$

where we have set:

$$\widetilde{R_n}^{4,1} = \int_{t_n}^{t_{n+1}}\int_{t_n}^{s}\int_{t_n}^{r}dudW_rdW_s \quad \text{and,} \quad \widetilde{R_n}^{4,2} = \int_{t_n}^{t_{n+1}}\int_{t_n}^{s}\int_{t_n}^{r}dW_udW_rdW_s. \tag{3.37}$$

We can now introduce a scheme that includes the $\mathcal{O}(\Delta t)$ error term and give us the so-called Milstein Scheme of strong order 1.

Definition 3.2.2 (The Milstein Scheme)**.** *For all $t_n < t < t_{n+1}$, the Milstein Scheme is given by:*

$$\begin{aligned} Y_{n+1} &= Y_n + b(Y_n,t_n)\Delta t + \sigma(Y_n,t_n)\Delta W_n + \sum_{k=1}^{m} \mathcal{L}^k_{t_n}\sigma(Y_n,t_n)\int_{t_n}^{t_{n+1}}\int_{t_n}^{s} dW_r^k \overrightarrow{dW_s} \\ &= Y_n + b(Y_n,t_n)\Delta t + \sigma(Y_n,t_n)\Delta W_n + \sum_{k=1}^{m}\sum_{l=1}^{m} \mathcal{L}^k_{t_n}\sigma^l(Y_n,t_n)\int_{t_n}^{t_{n+1}}\int_{t_n}^{s} dW_r^k dW_s^l, \end{aligned}$$

where we have σ^l as the l-th column of σ, also,

$$I_n(k,l) := \int_{t_n}^{t_{n+1}}\int_{t_n}^{s} dW_r^k dW_s^l = \int_{t_n}^{t_{n+1}} \left(W_s^k - W_{t_n}^k\right) dW_s^l, \tag{3.38}$$

and finally,

$$\mathcal{L}^k_{t_n}\sigma^l(Y_n,t_n) = \sigma^k(Y_n,t_n)\cdot\nabla\sigma^l(Y_n,t_n). \tag{3.39}$$

3.3 Properties of $I_n(k,l)$ and Derivative-Free Schemes

In general, it is quite difficult to compute the double stochastic integral (3.38). In this section we present some cases where computing $I_n(k,l)$ is computationally feasible and implementable.

- We start with the case where $k = l$. For $I_n(k,k)$ we use Itô's Formula and set $\widetilde{W_s}^k = W_s^k - W_{t_n}^k$ to find:

$$I_n(k,k) = \int_{t_n}^{t_{n+1}} \left(W_s^k - W_{t_n}^k\right) dW_s^k = \int_{t_n}^{t_{n+1}} \widetilde{W_s}^k dW_s^k = \int_{t_n}^{t_{n+1}} \widetilde{W_s}^k d\widetilde{W_s}^k. \tag{3.40}$$

Now we have that:

$$\begin{aligned} d\left(\left(\widetilde{W_s}^k\right)^2\right) = 2\widetilde{W_s}^k d\widetilde{W_s}^k + ds &= \frac{1}{2}\int_{t_n}^{t_{n+1}} d\left(\left(\widetilde{W_t}^k\right)^2\right) - \frac{1}{2}\Delta t \\ &= \frac{1}{2}\left(\widetilde{W}^k_{t_{n+1}}\right)^2 - \frac{1}{2}\Delta t \\ &= \frac{1}{2}\left(\Delta W_n^2\right)^2 - \frac{1}{2}\Delta t. \end{aligned}$$

Therefore, the SDE with $m = 1$ becomes:

$$Y_{n+1} = Y_n + b(Y_n,t_n)\Delta t + \sigma(Y_n,t_n)\Delta W_n + \sigma(Y_n,t_n)\cdot\nabla\sigma(Y_n,t_n)\frac{1}{2}\left(\Delta W_n^2 - \Delta t\right). \tag{3.41}$$

- We can also have an SDE with additive noise that is independent of X_t. In this case, the Milstein Scheme gives the Euler-Maruyama Scheme which is $\mathcal{O}(\Delta t)$, and hence has strong order 1.

- Another case is an SDE with diagonal noise:

$$\sigma(x,t) = \begin{pmatrix} \sigma^{(1)}(x_1,t) & & & \\ & \sigma^{(2)}(x_2,t) & & \\ & & \ddots & \\ & & & \sigma^{(d)}(x_d,t) \end{pmatrix}. \tag{3.42}$$

Then, $\mathcal{L}^k_{t_n}\sigma^l(Y_n,t_n)=0$ for $l\neq k$, since the l-th column of σ depends only on x_l, and hence:

$$Y_{n+1}=Y_n+b(Y_n,t_n)\Delta t+\sigma(Y_n,t_n)\Delta W_n+\sum_{k=1}^{m}\mathcal{L}^k_{t_n}\sigma^k(Y_n,t_n)I_n(k,k). \tag{3.43}$$

- We can further consider an SDE with commutative noise:

$$\mathcal{L}^k_{t_n}\sigma^l(Y_n,t_n)=\mathcal{L}^k_{t_n}\sigma^k(Y_n,t_n). \tag{3.44}$$

First note that $I_n(k,l)+I_n(l,k)=\Delta W^k_n\Delta W^l_n$. Indeed, along with $d(Z^k_tZ^l_t)=Z^k_tdZ^l_t+Z^l_tdZ^k_t+\underbrace{d\langle Z^k_t,Z^l_t\rangle_t}_{=0}$, it yields:

$$I_n(k,l)+I_n(l,k)=\int_{t_n}^{t_{n+1}}Z^k_tdZ^l_t+Z^l_tdZ^k_t=Z^k_{t_{n+1}}Z^l_{t_{n+1}}=\Delta W^k_n\Delta W^l_n. \tag{3.45}$$

If we now define $A_n(k,l)=I_n(k,l)-I_n(l,k)$ which are called the Lévy areas, we obtain that $I_n(k,l)=\frac{1}{2}\Delta W^k_n\Delta W^l_n+\frac{1}{2}A_n(k,l)$. Therefore, we can write the Milstein Scheme as:

$$\begin{aligned}
Y_{n+1}&=Y_n+b(Y_n,t_n)\Delta t+\sigma(Y_n,t_n)\Delta W_n+\frac{1}{2}\sum_{k,l=1}^{m}\mathcal{L}^k_{t_n}\sigma^l(Y_n,t_n)(\Delta W^k_n\Delta W^l_n-\Delta t\delta_{kl}+A_n(k,l))\\
&=Y_n+b(Y_n,t_n)\Delta t+\sigma(Y_n,t_n)\Delta W_n+\frac{1}{2}\sum_{k=1}^{m}\mathcal{L}^k_{t_n}\sigma^k(Y_n,t_n)\left(\left(\Delta W^k_n\right)^2-\Delta t\right)\\
&+\frac{1}{2}\sum_{k=1}^{m}\sum_{l<k}\left(\mathcal{L}^k_{t_n}\sigma^l(Y_n,t_n)+\mathcal{L}^l_{t_n}\sigma^k(Y_n,t_n)\right)\Delta W^k_n\Delta W^l_n+\frac{1}{2}\sum_{k=1}^{m}\sum_{l<k}\left(\mathcal{L}^k_{t_n}\sigma^l(Y_n,t_n)-\mathcal{L}^l_{t_n}\sigma^k(Y_n,t_n)\right)A_n(k,l).
\end{aligned}$$

The problematic term is the one involving the Lévy areas since there is no closed form expression for those, and they can only be simulated approximately by some series expansion. However, the truncation level in such an expansion has to be linked to Δt to recover the order 1 convergence and the overall cost turns out to be higher than that of the Euler-Maruyama Scheme with step size Δt^2.

- Finally, we present an example of a derivative-free scheme for an SDE with diagonal noise. Consider the deterministic interpolation $\overline{Y}^k_n$ given by:

$$\overline{Y}^k_n=Y_n+b(Y_n,t_n)\Delta t+\sigma^k(Y_n,t_n)\sqrt{\Delta t}. \tag{3.46}$$

Then, the SDE becomes:

$$Y_{n+1}=Y_n+b(Y_n,t_n)\Delta t+\sigma(Y_n,t_n)\Delta W_n+\sum_{k=1}^{m}\frac{\sigma^k(\overline{Y}^k_n,t_n)-\sigma^k(Y_n,t_n)}{\sqrt{\Delta t}}I_n(k,k). \tag{3.47}$$

3.4 Weak Convergence of SDE Schemes

In this section we uncover the notion of weak convergence through the analysis of the Euler-Maruyama Scheme, the Feynman-Kac Formula and using methods of Taylor estimates.

3.4.1 Weak Convergence of the Euler-Maruyama Scheme

For a d-dimensional SDE $dX_t = b(X_t,t)dt + \sigma(X_t,t)dW_t$ with initial condition $X_0 = \eta$ and m-dimensional Brownian motion, we recall the Euler-Maruyama Scheme with initial condition $Y_0 = \eta$:

$$Y_{n+1} = Y_n + b(Y_n,t_n)\Delta t + \sigma(Y_n,t_n)\Delta W_n, \tag{3.48}$$

and its interpolated version for $t \in (t_n, t_{n+1}]$:

$$\widehat{Y}_t = Y_n + b(Y_n,t_n)(t-t_n) + \sigma(Y_n,t_n)(W_t - W_{t_n}). \tag{3.49}$$

Given $\phi : \mathbb{R}^d \longrightarrow \mathbb{R}$ smooth with polynomial growth, the goal is to compute $I := \mathbb{E}(\phi(X_T))$. We also consider its approximation $\widehat{I} := \mathbb{E}(\phi(Y_N)) = \mathbb{E}\left(\phi(\widehat{Y_T})\right)$.

Theorem 3.4.1. *Consider $b, \sigma \in C^{1,2}$ with bounded derivatives. Suppose further that:*

- *b, σ satisfy a Linear Growth Bound,*
- *$\eta \in L^p(\Omega)$ for all p,*
- *$u \in C^{1,4}$ with polynomial growth, which implies that $\phi \in C^4$ with polynomial growth.*

Then $|I - \widehat{I}|$ corresponds to a sum of local errors of order $\mathcal{O}(\Delta t^2)$, and therefore:

$$|I - \widehat{I}| = \mathcal{O}(\Delta t). \tag{3.50}$$

First, for $t \in [0,T]$ recall the Backward Kolmogorov equation:

$$u(x,t) = \mathbb{E}\left[\phi(X_T^{x,t})\right], \tag{3.51}$$

where $u(x,t)$ is the solution of:

$$\begin{cases} \partial_t u(x,t) + \mathcal{L}_t u(x,t) = 0, & t \in [0,T), x \in \mathbb{R}^d \\ u(x,T) = \phi(x) \end{cases}. \tag{3.52}$$

We now introduce the setup for the proof of the above theorem. The error representation for the Euler-Maruyama Scheme can be derived as:

$$\begin{aligned} |I - \widehat{I}| &= \left|\mathbb{E}(\phi(X_T)) - \mathbb{E}\left(\phi(\widehat{Y}_T)\right)\right| \\ &= \left|\mathbb{E}\left[\mathbb{E}(\phi(X_T)|\mathcal{F}_0)\right] - \mathbb{E}\left[\mathbb{E}\left(\phi(\widehat{Y}_T)|\widehat{F}_T\right)\right]\right| \\ &= \left|\mathbb{E}(u(\eta,0)) - \mathbb{E}\left(u(\widehat{Y}_T,T)\right)\right|, \end{aligned}$$

where we had that $\mathbb{E}(u(\eta,0)) = \mathbb{E}\left(u(\widehat{Y}_0,0)\right)$. Now, using the properties of telescoping sums we have:

$$\begin{aligned} |I - \widehat{I}| &= |\mathbb{E}(u(Y_N,T)) - \mathbb{E}(u(Y_0,0))| \\ &= \left|\sum_{n=0}^{N-1} \mathbb{E}(u(Y_{n+1},t_{n+1})) - \mathbb{E}(u(Y_n,t_n))\right| \\ &\leq \sum_{n=0}^{N-1} \underbrace{|\mathbb{E}(u(Y_{n+1},t_{n+1})) - \mathbb{E}(u(Y_n,t_n))|}_{=\tau_n} =: \sum_{n=0}^{N-1} \tau_n. \end{aligned}$$

We denoted by τ_n the local weak error contributions, and the aim will be to show that these errors are $\mathcal{O}(\Delta t^2)$. Before starting the proof, we first provide two characterisations of τ_n.

- Considering that we are using a standard Brownian motion we have:

$$\mathbb{E}\left[u(Y_n,t_n)\right] = \mathbb{E}\left[\phi(X_T^{Y_n,t_n})\right] = \mathbb{E}\left[\mathbb{E}\left(\phi(X_T^{Y_n,t_n})|\mathcal{F}_{t_{n+1}}\right)\right] = \mathbb{E}\left[\mathbb{E}\left(X_T^{X_{t_{n+1}}^{Y_n,t_n},t_{n+1}}\right)\middle|\mathcal{F}_{t_{n+1}}\right] = \mathbb{E}\left[u\left(X_{t_{n+1}}^{Y_n,t_n},t_{n+1}\right)\right].$$

Therefore, τ_n can be expressed as:

$$\tau_n = \left|\mathbb{E}\left[u(Y_{n+1},t_{n+1}) - u\left(X_{t_{n+1}}^{Y_n,t_n},t_{n+1}\right)\right]\right|. \tag{3.53}$$

- Another characterisation of τ_n is:

$$\tau_n = \left|\mathbb{E}\left[u(\widehat{Y}_{t_{n+1}},t_{n+1}) - u(\widehat{Y}_{t_n},t_n)\right]\right|, \tag{3.54}$$

where, for $t \in (t_n, t_{n+1}]$, we consider the process,

$$d\widehat{Y}_t = b(Y_n,t_n)dt + \sigma(Y_n,t_n)dW_t. \tag{3.55}$$

Finally note that, given the diffusion matrix $\sigma\sigma^T$, the generator of u is:

$$\widehat{\mathcal{L}}_{t_n,Y_n}u = b(Y_n,t_n)\cdot\nabla u + \frac{1}{2}\sigma(Y_n,t_n)\sigma^T(Y_n,t_n) : \nabla^2 u. \tag{3.56}$$

We are now ready to present the proof of Theorem (3.4.1).

Proof. We use Itô's Formula on $u(\widehat{Y}_t,t)$ and find:

$$du(\widehat{Y}_t,t) = \left(\partial_t u(\widehat{Y}_t,t) + \widehat{\mathcal{L}}_{t_n,Y_n}u(\widehat{Y}_t,t)\right)dt + \sum_{k=1}^{m}\widehat{\mathcal{L}}^{(k)}_{Y_n,t_n}u(\widehat{Y}_t,t)dW_t^{(k)}, \tag{3.57}$$

where $\widehat{\mathcal{L}}^{(k)}_{Y_n,t_n} = \nabla^T u\cdot\overrightarrow{\sigma}^{(k)}(Y_n,t_n)$. Therefore τ_n can be written as:

$$\tau_n = \left|\mathbb{E}\left[\int_{t_n}^{t_{n+1}}\left(\partial_t u(\widehat{Y}_t,t) + \widehat{\mathcal{L}}_{t_n,Y_n}u(\widehat{Y}_t,t)\right)dt + \int_{t_n}^{t_{n+1}}\sum_{k=1}^{m}\widehat{\mathcal{L}}^{(k)}_{Y_n,t_n}u(\widehat{Y}_t,t)dW_t^{(k)}\right]\right|. \tag{3.58}$$

Now since,

$$\sum_{k=1}^{m}\widehat{\mathcal{L}}^{(k)}_{Y_n,t_n}u(\widehat{Y}_t,t) \le C(1+|\widehat{Y}_t|^l) \in M^2([t_n,t_{n+1}]), \tag{3.59}$$

then we have a martingale, which implies that the expectation value of the stochastic integral is equal to 0. Furthermore, using the fact that $\partial_t u(\widehat{Y}_t,t) = -\mathcal{L}_t u(\widehat{Y}_t,t)$ we obtain:

$$\begin{aligned}
\tau_n &= \left|\mathbb{E}\left[\int_{t_n}^{t_{n+1}}\left(\partial_t u(\widehat{Y}_t,t) + \widehat{\mathcal{L}}_{t_n,Y_n}u(\widehat{Y}_t,t)\right)dt\right]\right| \\
&= \left|\mathbb{E}\left[\int_{t_n}^{t_{n+1}}\left(\mathcal{L}_t u(\widehat{Y}_t,t) - \widehat{\mathcal{L}}_{t_n,Y_n}u(\widehat{Y}_t,t)\right)dt\right]\right| \\
&= \left|\mathbb{E}\left[\int_{t_n}^{t_{n+1}}(b(\widehat{Y}_t,t)-b(Y_n,t_n))\cdot\nabla u(\widehat{Y}_t,t)dt + \int_{t_n}^{t_{n+1}}\frac{1}{2}(A(\widehat{Y}_t,t)-A(Y_n,t_n)) : \nabla^2 u(\widehat{Y}_t,t)dt\right]\right|.
\end{aligned}$$

We denote the first integral by A, the second by B, and denote $g(\widehat{Y}_t,t) = (b(\widehat{Y}_t,t)-b(Y_n,t_n))\cdot$

$\nabla u(\widehat{Y}_t, t)$. We again use Itô's Formula on the term A and we find:

$$\begin{aligned} A &= \left| \mathbb{E}\left[\int_{t_n}^{t_{n+1}} g(\widehat{Y}_t, t) dt \right] \right| \\ &= \left| \mathbb{E}\left[\int_{t_n}^{t_{n+1}} \left(\cancelto{0}{g(\widehat{Y}_n, t_n)} + \int_{t_n}^{t} (\partial_t g + \widehat{\mathcal{L}}_{t_n, Y_n} g) dt + ... \right) \right] \right| \\ &= \left| \mathbb{E}\left[\int_{t_n}^{t_{n+1}} \int_{t_n}^{t} (\partial_t g + \widehat{\mathcal{L}}_{t_n, Y_n} g) dt \right] \right| = \mathcal{O}(\Delta t^2). \end{aligned}$$

We can similarly show that $B = \mathcal{O}(\Delta t^2)$ and summing over $n = \{0, ..., N-1\}$ yields the result.

□

3.4.2 Weak Convergence with the setting of Feynman-Kac

We consider a d-dimensional SDE:

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dW_t, \tag{3.60}$$

and the Euler-Maruyama Scheme:

$$Y_{n+1} = Y_n + b(Y_n, t_n)\Delta t + \sigma(Y_n, t_n)\Delta \widehat{W}_n. \tag{3.61}$$

Given $\phi : \mathbb{R}^d \to \mathbb{R}$ smooth and with polynomial growth, we consider the error representation:

$$\mathbb{E}\left(\phi(X_T)\right) - \mathbb{E}\left(\phi(Y_N)\right) = \sum_{n=0}^{N-1} \mathbb{E}\left[u(Y_n, t_n) - u(Y_{n+1}, t_{n+1})\right], \tag{3.62}$$

where u solves the following system for $x \in \mathbb{R}^d$ and $t \in [0, T)$:

$$\begin{cases} \partial_t u + \mathcal{L}_t u = 0 \\ u(x, T) = \phi(x) \end{cases}. \tag{3.63}$$

In this section we consider the setup of Feynman-Kac's Formula [Theorem (2.4.3)]. We then take this generalization and compute the error of:

$$I = \mathbb{E}\left[\phi(X_T) e^{-\int_0^T c(X_t, t)dt} - \int_0^T f(X_t, t) e^{-\int_0^t c(X_s, s)ds} dt \right], \tag{3.64}$$

where we recall that:

$$u(x, t) = \mathbb{E}\left[\phi\left(X_T^{x,t}\right) e^{-\int_t^T c(X_s^{x,t}, s)ds} \right] - \mathbb{E}\left[\int_t^T f(X_v^{x,t}, v) e^{-\int_t^s c(X_w^{x,t}, w)dw} dv \right], \tag{3.65}$$

and as a result, for $(x, t) \in \mathbb{R}^d \times [0, T)$, $u(x, t)$ solves:

$$\begin{cases} \partial_t u + \mathcal{L}_t u - cu = f \\ u(x, T) = \phi(x) \end{cases}. \tag{3.66}$$

We now consider the numerical estimation given by:

$$\widehat{I} = \mathbb{E}\left[\phi(Y_N)e^{-\sum_{n=0}^{N-1} c(Y_n,t_n)\Delta t} - \sum_{n=0}^{N-1} f(Y_n,t_n)e^{-\sum_{j=0}^{N-1} c(Y_j,t_j)\Delta t}\Delta t\right]. \tag{3.67}$$

Introducing,

$$\widehat{Y}_s = Y_n + b(Y_n,t_n)(s-t_n) + \sigma(Y_n,t_n)(W_s - W_{t_n}), \tag{3.68}$$

we can rewrite this estimation as:

$$\widehat{I} = \mathbb{E}\left[\phi(Y_N)e^{-\int_0^T c(\widehat{Y}_s,\phi_s)ds} - \int_0^T f(\widehat{Y}_{n_t},\phi_t)e^{-\int_0^{\phi_t} c(\widehat{Y}_s,\phi_s)ds}dt\right]. \tag{3.69}$$

For simplicity we introduce the following notations:

$$z_t = e^{-\int_0^t c(X_s,s)ds} \quad \text{and} \quad \widehat{z}_t = e^{-\int_0^t c(\widehat{Y}_s,\phi_s)ds}. \tag{3.70}$$

Theorem 3.4.2. *Consider $b,\sigma,f,c \in C^{2,1}$ with derivatives growing at most polynomially. Suppose further that:*

- *c is bounded from below,*
- *$u \in C^{4,1}$ with polynomial growth.*

Then:

$$|I - \widehat{I}| = \mathcal{O}(\Delta t). \tag{3.71}$$

Proof. We have:

$$\begin{aligned}|\widehat{I} - I| &= \mathbb{E}\left[\phi(\widehat{Y}_T)\widehat{z}_T - \int_0^T f(\widehat{Y}_{n_t},\phi_t)\widehat{z}_{\phi_t}dt - u(X_0,0)\right] \\ &= \mathbb{E}\left[u(\widehat{Y}_T,T)\widehat{z}_T - u(\widehat{Y}_0,0)\widehat{z}_0 - \int_0^T f(\widehat{Y}_{n_t},\phi_t)\widehat{z}_{\phi_t}dt\right].\end{aligned}$$

For $t \in [t_n,t_{n+1}]$, the generator is given by:

$$\widehat{\mathcal{L}}_t = b(Y_n,t_n)\cdot\nabla + \frac{1}{2}\sigma(Y_n,t_n)\sigma^T(Y_n,t_n) : \nabla^2. \tag{3.72}$$

Therefore, we can simplify $|\widehat{I} - I|$ further to find:

$$\begin{aligned}|\widehat{I} - I| = \mathbb{E}\Bigg[\int_0^T \left(\partial_t u(\widehat{Y}_t,t)\widehat{z}_t + \widehat{\mathcal{L}}_t u(\widehat{Y}_t,t)\widehat{z}_t - c(\widehat{Y}_{n_t},\phi_t)\widehat{z}_t u(\widehat{Y}_t,t)\right)dt \\ + \underbrace{\int_0^T \nabla^T u(\widehat{Y}_t,t)\sigma(\widehat{Y}_t,t)d\widehat{W}_t}_{=0 \text{ under usual assumptions}} - \int_0^T f(\widehat{Y}_{n_t},\phi_t)\widehat{z}_t dt\Bigg].\end{aligned}$$

Remark that $\widehat{\mathcal{L}}_t$ is the approximated generator of the interpolated Euler-Maruyama process. We need to recover the Backward Kolmogorov equation, hence we have to introduce the true generator

$\pm\mathcal{L}_t u(\widehat{Y}_t,t)$ but also $\pm c(\widehat{Y}_t,\phi_t)$ and $\pm f(\widehat{Y}_t,t)\widehat{z}_t$. Hence we obtain:

$$\begin{aligned}|\widehat{I}-I| &= \mathbb{E}\Big[(\widehat{\mathcal{L}}_t-\mathcal{L}_t)u(\widehat{Y}_t,t)\widehat{z}_t+(c(\widehat{Y}_t,t)-c(\widehat{Y}_{n_t},\phi_t))u(\widehat{Y}_t,t)\widehat{z}_t+(\partial_t+\mathcal{L}_t-c)u(\widehat{Y}_t,t)\widehat{z}_t\\ &\qquad -f(\widehat{Y}_t,t)\widehat{z}_t+(f(\widehat{Y}_t,t)\widehat{z}_t-f(\widehat{Y}_{n_t},\phi_t)\widehat{z}_{\phi_t})\Big]\\ &= \ldots \text{ (Itô's Formula)}\\ &= \mathcal{O}(\Delta t).\end{aligned}$$

□

3.4.3 Weak Convergence Using Taylor Estimates

In this subsection we go back to $I=\mathbb{E}(\phi(X_T))\approx\widehat{I}=\mathbb{E}(\phi(Y_N))$. Recall that we have the following upper-bound:

$$\begin{aligned}|I-\widehat{I}| = \mathbb{E}(u(Y_0,0)-u(Y_N,T)) &= \left|\mathbb{E}\left[\sum_{n=0}^{N-1}(u(Y_n,t_n)-u(Y_{n+1},t_{n+1}))\right]\right|\\ &\le \sum_{n=0}^{N-1}|\mathbb{E}[u(Y_n,t_n)-u(Y_{n+1},t_{n+1})]| =: \sum_{n=0}^{N-1}\tau_n.\end{aligned}$$

Not all processes can be interpolated, hence one could use error estimates using Taylor expansions instead of Itô. Since the calculations become long and tedious, we only look at dimensions $d=1$ and $m=1$. We have:

$$\tau_n = |\mathbb{E}(u(Y_{n+1},t_{n+1})-u(Y_n,t_n))| = \left|\mathbb{E}\left(T_1^{(Y_n,t_n)}+T_2^{(Y_n,t_n)}+T_3^{(Y_n,t_n)}+R\right)\right|. \tag{3.73}$$

Further let $u_n := u(Y_n,t_n)$, $b_n := b(Y_n,t_n)$ and $\sigma_n := \sigma(Y_n,t_n)$. We now expand to order 3 and regroup terms by their order i of derivatives, denoting them by T_i.

- For T_1 we find:

$$T_1 = \partial_t u_n\Delta t+\partial_x u_n\Delta Y_n = \partial_t u_n\Delta t+\partial_x u_n(b_n\Delta t+\underbrace{\sigma_n\Delta W_n}_{\mathbb{E}=0}).$$

- For T_2 we find:

$$\begin{aligned}T_2 &= \frac{1}{2}\sigma_{tt}^2u_n\Delta t^2+\partial_{tx}^2u_n\Delta t\Delta Y_n+\frac{1}{2}\partial_{xx}^2u_n(\Delta Y_n)^2\\ &= \frac{1}{2}\sigma_{tt}^2u_n\Delta t^2+\partial_{tx}^2u_nb_n\Delta t^2+\underbrace{\partial_{tx}^2u_n\sigma_n\Delta t\Delta W_n}_{\mathbb{E}=0}+\frac{1}{2}\partial_{xx}^2u_nb_n^2\Delta t^2+\frac{1}{2}\partial_{xx}^2u_n\sigma_n^2\underbrace{\Delta W_n^2}_{\mathbb{E}=\Delta t}+\underbrace{\partial_{xx}^2u_nb_n\sigma_n\Delta t\Delta W_n}_{\mathbb{E}=0}.\end{aligned}$$

- Finally, for T_3 we find:

$$\begin{aligned}T_3 &= \underbrace{\frac{1}{6}\partial_{ttt}^3u_n\Delta t^3}_{\text{Higher Order Term}}+\underbrace{\frac{1}{2}\partial_{ttx}^3u_n\Delta t^2\Delta Y_n}_{\text{Higher Order Term}}+\frac{1}{2}\partial_{txx}^3u_n\Delta t(\Delta Y_n)^2+\frac{1}{6}\partial_{xxx}^3u_n(\Delta Y_n)^3\\ &= \frac{1}{2}\partial_{txx}^3u_n\sigma_n^2\Delta t\underbrace{\Delta W_n^2}_{\mathbb{E}=\Delta t}+\frac{1}{6}\partial_{xxx}^3u_n(3b_n)\sigma_n^2\underbrace{\Delta W_n^2}_{\mathbb{E}=\Delta t}\Delta t.\end{aligned}$$

Moreover, the remainder is $R = \mathcal{O}(\Delta t^2)$. Therefore:

$$\tau_n = |\mathbb{E}(T_1 + T_2 + T_3 + R)| = \Big| \Big(\underbrace{\partial_t u_n + b_n \partial_x u_n + \frac{1}{2}\sigma_n^2 \partial_{xx}^2 u_n}_{(\partial_t + \mathcal{L}_t)u_n = 0} \Big) \Delta t + \mathcal{O}(\Delta t^2) \Big| = \mathcal{O}(\Delta t^2). \tag{3.74}$$

Hence, $|I - \widehat{I}| = \sum_{n=0}^{N-1} \tau_n = \mathcal{O}(\Delta t)$. In the above calculations, we have used the facts that $\mathbb{E}\,(\Delta W_n | \mathcal{F}_n) = 0$, $\mathbb{E}\,(\Delta(W_n)^2 | \mathcal{F}_n) = \Delta t$, $\mathbb{E}\,(\Delta(W_n)^3 | \mathcal{F}_n) = 0$ and $\mathbb{E}\,(\Delta(W_n)^4 | \mathcal{F}_n) = \mathcal{O}(\Delta t^2)$. Note that at no point in the analysis have we used the fact that the Brownian Motion W_t is a Gaussian Process, nor have we used Itô's Formula. In fact, we only had to use the four facts given previously.

3.5 Higher Order Schemes

To treat higher order schemes we can use an Itô Taylor expansion. Set,

$$X_{t_{n+1}} = X_{t_n} + \int_{t_n}^{t_{n+1}} b(X_t, t)dt + \int_{t_n}^{t_{n+1}} \sigma(X_t, t) dW_t. \tag{3.75}$$

Now, to continue the analysis further we can add and substract $\pm b(X_{t_n}, t_n)$ and $\pm\sigma(X_{t_n}, t_n)$, use Itô's Formula to find:

$$\begin{aligned}
X_{t_{n+1}} &= \underbrace{X_{t_n} + b(X_{t_n}, t_n)\Delta t + \sigma(X_{t_n}, t_n)\Delta W_n}_{=\text{EM}} + \int_{t_n}^{t_{n+1}} (b(X_t, t) - b(X_{t_n}, t_n))dt \\
&\qquad + \int_{t_n}^{t_{n+1}} (\sigma(X_t, t) - \sigma(X_{t_n}, t_n))dW_t \\
&= \text{EM} + \int_{t_n}^{t_{n+1}} \int_{t_n}^{t} \left((\partial_s + \mathcal{L}_s) b(X_s, s) ds + \sum_{k=1}^{m} \mathcal{L}_s^k b(X_s, s) dW_s^k \right) dt \\
&\qquad + \int_{t_n}^{t_{n+1}} \int_{t_n}^{s} \left((\partial_s + \mathcal{L}_s)\sigma(X_s, s) ds + \sum_{k=1}^{m} \mathcal{L}_s^k \sigma(X_s, s) dW_s^k \right) dW_t \\
&= \text{EM} + (\partial_{t_n} + \mathcal{L}_{t_n})\, b(X_{t_n}, t_n) \left(\int_{t_n}^{t_{n+1}} \int_{t_n}^{t} ds dt \right) + \sum_{k=1}^{m} \mathcal{L}_{t_n}^k b(X_{t_n}, t_n) \left(\int_{t_n}^{t_{n+1}} \int_{t_n}^{t} dW_s^k dt \right) \\
&\quad + \sum_{k=1}^{m} (\partial_{t_n} + \mathcal{L}_{t_n})\, \sigma^k(X_{t_n}, t_n) \left(\int_{t_n}^{t_{n+1}} \int_{t_n}^{t} ds dW_t^k \right) + \sum_{k,k'=1}^{m} \mathcal{L}_{t_n}^k \sigma^{k'}(X_{t_n}, t_n) \left(\int_{t_n}^{t_{n+1}} \int_{t_n}^{t} dW_s^k dW_t^{k'} \right) + \text{Rem.}
\end{aligned}$$

This previous derivation gives us the Weak Milstein Scheme, and it can be written as the integrals $I_n(k,l) := \int_{t_n}^{t_{n+1}} \int_{t_n}^{t} dW_s^k dW_t^l$:

$$\begin{aligned}
Y_{n+1} = Y_n &+ b(Y_n, t_n)\Delta t + \sigma(Y_n, t_n)\Delta W_n + (\partial_{t_n} + \mathcal{L}_{t_n})\, b(X_{t_n}, t_n) I_n(0,0) + \sum_{k=1}^{m} \mathcal{L}_{t_n}^k b(X_{t_n}, t_n) I_n(k,0) \\
&+ \sum_{k=1}^{m} (\partial_{t_n} + \mathcal{L}_{t_n})\, \sigma^k(X_{t_n}, t_n) I_n(0,k) + \sum_{k,k'=1}^{m} \mathcal{L}_{t_n}^k \sigma^{k'}(X_{t_n}, t_n) I_n(k,k') + \text{Remainders.}
\end{aligned}$$

This is a valid scheme, however it involves a lot of derivatives of b and σ. Furthermore, we can only solve this scheme for particular cases where we can explicitly find the stochastic integrals $I_n(k,l)$, since we have seen in general that there are no closed form formulas for them.

3.6 Asymptotic Stability

In this section, we consider a d-dimensional SDE with m-dimensional noise:

$$\begin{cases} dX_t = b(X_t,t)dt + \sigma(X_t,t)dW_t \\ X_{t_0} = x_0 \end{cases}. \tag{3.76}$$

Suppose that $X_t^{t_0,x_0} = 0$ is a solution, which would imply that $b(0,t) = \sigma(0,t) = 0$. We are interested to know the cases where $X_t^{t_0,x_0} \longrightarrow 0$ as $n \to \infty$.

3.6.1 Asymptotic Stability of SDEs

We start with a list of definitions to characterize some asymptotic results for solutions of SDEs.

Definition 3.6.1. *We say that $X_t = 0$ is **stochastically asymptotically stable** if $\forall \epsilon \in (0,1)$, $\exists \delta = \delta(\epsilon, t_0) > 0$ such that $\forall |x_0| < \delta$ one has:*

$$\mathbb{P}\left(\lim_{t\to\infty} X_t^{t_0,x_0} = 0\right) \geq 1 - \epsilon. \tag{3.77}$$

Definition 3.6.2. *We say that $X_t = 0$ is **stochastically asymptotically stable in the large** (or Globally Stable) if $\forall x_0 \in \mathbb{R}^d$:*

$$\lim_{t\to\infty} X_t^{t_0,x_0} = 0 \ \ a.s. \tag{3.78}$$

Definition 3.6.3. *We say that $X_t = 0$ is **exponentially asymptotically stable** if:*

$$\limsup_{t\to\infty} \frac{1}{t} \log\left(\left|X_t^{t_0,x_0}\right|\right) < 0 \ \ a.s. \qquad (i.e.\ \left|X_t^{t_0,x_0}\right| \leq Ce^{-\alpha t}) \tag{3.79}$$

Definition 3.6.4. *We say that $X_t = 0$ is **mean-square asymptotically stable** if:*

$$\lim_{t\to\infty} \mathbb{E}\left[\left|X_t^{t_0,x_0}\right|^2\right] = 0. \tag{3.80}$$

Definition 3.6.5. *We say that $X_t = 0$ is **exponentially mean-square asymptotically stable** if:*

$$\limsup_{t\to\infty} \frac{1}{t} \log \mathbb{E}\left[\left|X_t^{t_0,x_0}\right|^2\right] < 0. \tag{3.81}$$

To illustrate the different notions of stability, we use a Geometric Brownian Motion to see how they can be shown, especially proving asymptotic and mean-square stability.

Example 3.6.6 (Geometric Brownian Motion)**.** *Consider the GBM given by:*

$$\begin{cases} dX_t = \mu X_t dt + \sum_{k=1}^m \sigma_k X_t dW_t^{(k)}, \ t \geq 0 \\ X_{t_0} = x_0 > 0 \end{cases} . \tag{3.82}$$

As usual, set $Y_t = \log(X_t)$ and use Itô's Formula to find:

$$X_t = x_0 \exp\left\{ \left(\mu - \frac{1}{2} \sum_{k=1}^m \sigma_k^2 \right)(t - t_0) + \sum_{k=1}^m \sigma_k \left(W_t^{(k)} - W_{t_0}^{(k)} \right) \right\}. \tag{3.83}$$

- *Asymptotic Stability*

Applying the definition we have:

$$\frac{1}{t} \log \left| X_t^{t_0, x_0} \right| = \underbrace{\frac{\log|x_0|}{t}}_{\to 0} + \underbrace{\frac{1}{t} \left(\mu - \frac{1}{2} \sum_{k=1}^m \sigma_k^2 \right)(t - t_0)}_{\to \mu - \frac{1}{2}\sum_k \sigma_k^2} + \underbrace{\frac{1}{t} \sum_{k=1}^m \sigma_k \left(W_t^{(k)} - W_{t_0}^{(k)} \right)}_{\to 0 \ a.s.}. \tag{3.84}$$

We have treated the last sum using the Law of Iterated Logarithms. Indeed, since $\limsup \frac{W_t}{\sqrt{2t \log\log(t)}} = 1$ and $\liminf \frac{W_t}{\sqrt{2t \log\log(t)}} = -1$, we have that $-\sqrt{2t \log\log(t)} \leq W_t \leq \sqrt{2t \log\log(t)}$ and hence yields the result. Overall:

$$\lim_{t \to 0} \frac{1}{t} \log \left| X_t^{t_0, x_0} \right| = \mu - \frac{1}{2} \sum_k \sigma_k^2. \tag{3.85}$$

Furthermore, the SDE is exponentially asymptotically stable if $\mu < \frac{1}{2} \sum_k \sigma_k^2$.

- *Mean-Square Stability*

Applying the definition and recalling that $W_t^{(k)} - W_{t_0}^{(k)} \sim \mathcal{N}(0, t - t_0)$, we have:

$$\begin{aligned} \mathbb{E}\left[\left| X_t^{t_0, x_0} \right|^2 \right] &= x_0^2 \exp\left\{ 2 \left(\mu - \frac{1}{2} \sum_k \sigma_k^2 \right)(t - t_0) \right\} \mathbb{E}\left[\exp\left\{ 2 \sum_k \sigma_k^2 \left(W_t^{(k)} - W_{t_0}^{(k)} \right) \right\} \right] \\ &= x_0^2 \exp\left\{ 2 \left(\mu - \frac{1}{2} \sum_k \sigma_k^2 \right)(t - t_0) \right\} \prod_{k=1}^m \mathbb{E}\left[\exp\left\{ 2\sigma_k^2 \left(W_t^{(k)} - W_{t_0}^{(k)} \right) \right\} \right] \\ &= x_0^2 \exp\left\{ 2 \left(\mu - \frac{1}{2} \sum_k \sigma_k^2 \right)(t - t_0) \right\} \prod_{k=1}^m \exp\left(\frac{1}{2} (2\sigma_k)^2 (t - t_0) \right) \\ &= x_0^2 \exp\left\{ 2 \left(\mu + \frac{1}{2} \sum_k \sigma_k^2 \right)(t - t_0) \right\}. \end{aligned}$$

The SDE is Mean-Square stable if $\mu + \frac{1}{2} \sum_k \sigma_k^2 < 0$.

We end this section by presenting two criteria for stability using Lyapunov functions. In the following theorems, $\mathcal{L}_t$ is defined as in Proposition (2.3.4).

Theorem 3.6.7. *Assume that there exists a Lyapunov function $V \in C^{2,1}(\mathbb{R}^d \times [t_0, \infty); \mathbb{R}_+)$ such that:*

- $V(x,t) \geq C_1|x|^p$,
- $\mathcal{L}_t V(x,t) \leq C_2 V(x,t)$,
- $|b^T(x,t)\nabla_x V(x,t)| \geq C_3 V(x,t)$.

Then:

$$\limsup_{t\to\infty} \frac{1}{t} \log\left|X_t^{t_0,x_0}\right| \leq -\frac{C_3^2 - 2C_2}{2p} \quad a.s. \tag{3.86}$$

which implies that $X_t = 0$ is asymptotically stable if $C_3^2 > 2C_2$.

Theorem 3.6.8. *Assume that there exists a Lyapunov function $V \in C^{2,1}(\mathbb{R}^d \times [t_0, \infty); \mathbb{R}_+)$ such that:*

- $C_1|x|^p \leq V(x,t) \leq \widehat{C_1}|x|^p$,
- $\mathcal{L}_t V(x,t) \leq -C_2 V(x,t)$, $C_2 > 0$.

Then, we have p-th moment exponential stability:

$$\mathbb{E}\left[\left|X_t^{t_0,x_0}\right|^p\right] \leq \frac{C_2}{\widehat{C_1}}|x_0|^p e^{-C_2(t-t_0)}. \tag{3.87}$$

3.6.2 Asymptotic Stability of Numerical Methods

Let $\{Y_n\}_n$ be the discrete solution given by a numerical scheme. We would like a similar notion of asymptotic stability,

$$\lim_{n\to\infty} \frac{1}{n} \log(|Y_n|) = 0 \tag{3.88}$$

and mean-square stability,

$$\lim_{n\to\infty} \mathbb{E}\left[|Y_n|^2\right] = 0. \tag{3.89}$$

We consider the model SDE in 1-dimension $dX_t = \mu X_t dt + \sigma X_t dW_t$. We recall that we had an asymptotically stable solution if $\mu - \frac{1}{2}\sigma^2 < 0$ (unstable if > 0) and mean-square stable if $\mu + \frac{1}{2}\sigma^2 < 0$ (unstable if > 0). We use the stochastic θ-method:

$$Y_{n+1} = Y_n + \theta\mu Y_{n+1}\Delta t + (1-\theta)\mu Y_n \Delta t + \sigma Y_n \Delta W_n \Longrightarrow Y_{n+1} = \left(\frac{1 + (1-\theta)\mu\Delta t + \sigma\Delta W_n}{1 - \theta\mu\Delta t}\right) Y_n. \tag{3.90}$$

Study of Mean-Square Stability

We apply the definition to have:

$$\begin{aligned}
\mathbb{E}\left[|Y_{n+1}|^2\right] &= \frac{1}{(1-\theta\mu\Delta t)^2}\mathbb{E}\left[(1 + (1-\theta)\mu\Delta t + \sigma\Delta W_n)^2 Y_n^2\right] \\
&= \frac{1}{(1-\theta\mu\Delta t)^2}\mathbb{E}\left[(1 + (1-\theta)\mu\Delta t + \sigma\Delta W_n)^2\right]\mathbb{E}\left[|Y_n|^2\right] \\
&= \left(\frac{(1 + (1-\theta)\mu\Delta t)^2 + \sigma^2\Delta t}{(1-\theta\mu\Delta t)^2}\right)\mathbb{E}\left[|Y_n|^2\right].
\end{aligned}$$

We have mean-square stability if $\frac{(1+(1-\theta)\mu\Delta t)^2 + \sigma^2\Delta t}{(1-\theta\mu\Delta t)^2} < 1$. Furthermore:

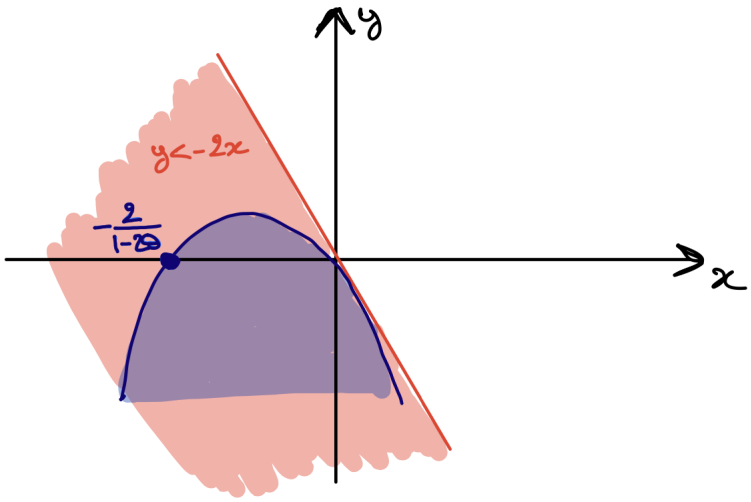
$$(1 + (1-\theta)\mu\Delta t)^2 - (1 - \theta\mu\Delta t)^2 < -\sigma^2\Delta t$$

$$\implies \mu(2+(1-2\theta)\mu\Delta t) < -\sigma^2$$
$$\implies (1-2\theta)\mu^2\Delta t < -\sigma^2 - 2\mu = -2\left(\mu + \frac{1}{2}\sigma^2\right).$$

- For $\theta \in [0, \frac{1}{2})$, $\Delta t < -\frac{2\left(\mu+\frac{1}{2}\sigma^2\right)}{(1-2\theta)\mu^2}$. Now, if $\mu + \frac{1}{2}\sigma^2 < 0$ (i.e. the SDE is mean-square stable) then the θ-method is mean-square stable if $\Delta t < \frac{2|\mu+\frac{1}{2}\sigma^2|}{(1-2\theta)\mu^2}$. Otherwise, the SDE is mean-square unstable implying that the θ-method is mean-square unstable for all Δt. Let $x = \Delta t\mu$ and $y = \Delta t\sigma^2$.

→ SDE stability region: $y < -2x$.

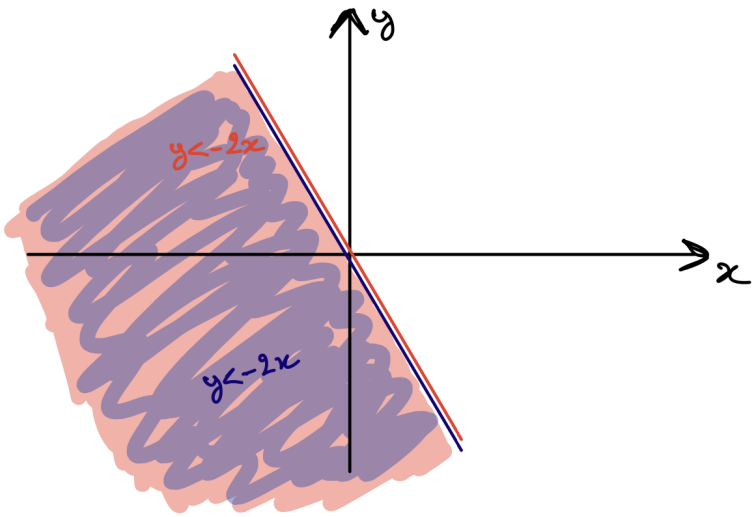
→ θ-method stability region: $y < -(1-2\theta)x^2 - 2x$.



- For $\theta = \frac{1}{2}$, if the SDE is mean-square stable then the θ-method is mean-square stable for all Δt. Otherwise, the SDE is mean-square unstable implying that the θ-method is mean-square unstable for all Δt.

→ SDE stability region: $y < -2x$.

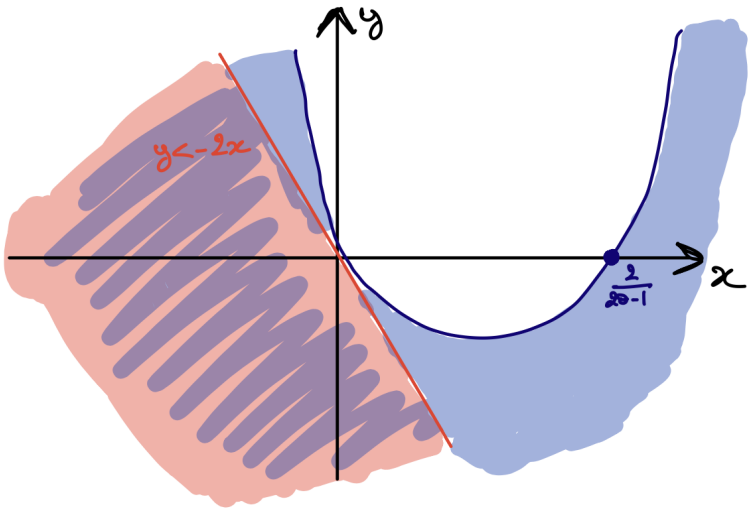
→ θ-method stability region: $y < -2x$.



- For $\theta \in (\frac{1}{2}, 1]$, $\Delta t > -\frac{2\left(\mu+\frac{1}{2}\sigma^2\right)}{(1-2\theta)\mu^2}$. Now, if the SDE is mean-square stable then the θ-method is mean-square stable for all Δt. Otherwise, the SDE is mean-square unstable implying that the θ-method is mean-square stable if $\Delta t > \frac{2|\mu+\frac{1}{2}\sigma^2|}{(1-2\theta)\mu^2}$.

→ SDE stability region: $y < -2x$.

→ θ-method stability region: $y < (2\theta-1)x^2 - 2x$.



Asymptotic Stability

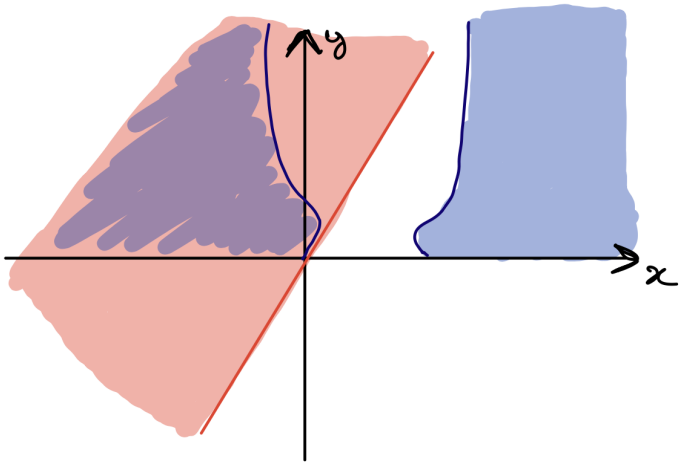
We recall the scheme $Y_{n+1} = \frac{1+(1-\theta)\mu\Delta t+\sigma\Delta W_n}{1-\theta\mu\Delta t}Y_n = V_nY_n$, with $V_n \sim \mathcal{N}\left(\frac{1+(1-\theta)\mu\Delta t}{1-\theta\mu\Delta t}, \frac{\sigma^2\Delta t}{(1-\theta\mu\Delta t)^2}\right)$. Hence, $Y_n = \left(\prod_{i=0}^{n-1} V_i\right)x_0$. Taking logarithms yields $\log|Y_n| = \log|x_0| + \sum_{i=0}^{n-1}\log|V_i|$ and set $Z_i = \log|V_i|$. Since Z_i are iid for $i = \{0,...,n-1\}$ with $\mathbb{E}[|Z_i|] < \infty$, by the Strong Law of Large Numbers we have $\frac{1}{n}\sum_i Z_i \longrightarrow \mathbb{E}[Z_1]$ a.s. and,

$$\lim_{n\to\infty}\frac{1}{n}\log|Y_n| = \lim_{n\to\infty}\left(\frac{1}{n}\log|x_0| + \frac{1}{n}\sum_{i=0}^{n-1}\log|V_i|\right) = \mathbb{E}[Z_1] \text{ a.s.} \tag{3.91}$$

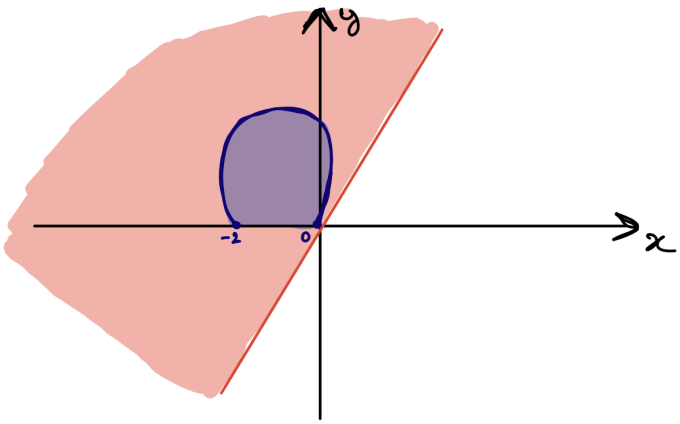
- If $\mathbb{E}[Z_1] = \mathbb{E}\left[\log\left|\frac{1+(1-\theta)\mu\Delta t+\sigma\Delta W_n}{1-\theta\mu\Delta t}\right|\right] < 0$ then $Y_n \to 0$ a.s. and therefore the θ-method is asymptotically stable.
- If $\mathbb{E}[Z_1] > 0$ then $\lim_n|Y_n| = \infty$ and the θ-method is asymptotically unstable.

Remark that the stability region $S = \left\{(x,y) : \mathbb{E}\left[\log\left|\frac{1+(1-\theta)x+\sqrt{y}\xi}{1-\theta x}\right|\right] < 0, \xi \sim \mathcal{N}(0,1)\right\}$ can be computed numerically.

→ For $\theta = 1$:



→ For $\theta = 0$:



3.7 Monte Carlo Simulation

In this section, we consider the following SDE:

$$\begin{cases} dX_t = b(X_t,t)dt + \sigma(X_t,t)dW_t, \quad t \in (0,T] \\ X_0 = \eta \end{cases} . \tag{3.92}$$

The goal is to compute the expectation $\mu = \mathbb{E}[\phi(X_T)]$ where $\phi : \mathbb{R}^d \longrightarrow \mathbb{R}$ is sufficiently smooth. Furthermore, let $\{Y_n\}_{n=0}^N$ be the solution produced by a numerical scheme with $\Delta t = \frac{T}{N}$ and we consider the usual errors:

$$\underline{\text{Weak Error}}: \quad |\mathbb{E}[\phi(X_T) - \phi(Y_n)]| = \mathcal{O}(\Delta t^{\alpha}) \tag{3.93}$$

$$\underline{\text{Strong Error}}: \quad \mathbb{E}\left[\sup_n |X_{t_n} - Y_n|^2\right] = \mathcal{O}(\Delta t^{2q}) \tag{3.94}$$

If we consider $\mu_{\Delta t} = \mathbb{E}[\phi(Y_n)]$ we can define a corresponding estimator.

Definition 3.7.1. *We define the* ***Monte Carlo estimator*** *as:*

$$\widehat{\mu}_{\Delta t,M} = \frac{1}{M}\sum_{i=1}^{M} \phi\left(Y_N^{(i)}\right), \quad Y_N^{(i)} \overset{\text{iid}}{\sim} Y_N. \tag{3.95}$$

We note that $\mathbb{E}\left(\widehat{\mu}_{\Delta t,M}\right) = \mu_{\Delta t}$, and we can now perform a mean-squared error analysis.

Proposition 3.7.2. *The Mean-Squared Error (MSE) is given by:*

$$\text{MSE}\left(\widehat{\mu}_{\Delta t,M}\right) := \mathbb{E}\left[\left(\widehat{\mu}_{\Delta t,M} - \mu\right)^2\right] = \mathcal{O}\left(\frac{1}{M}\right) + \mathcal{O}\left(\Delta t^{2\alpha}\right). \tag{3.96}$$

Proof. We first remark that:

$$\begin{aligned} \text{MSE}\left(\widehat{\mu}_{\Delta t,M}\right) := \mathbb{E}\left[\left(\widehat{\mu}_{\Delta t,M} - \mu\right)^2\right] &= \mathbb{E}\left[\left(\left(\widehat{\mu}_{\Delta t,M} - \mu_{\Delta t}\right) + \left(\mu_{\Delta t} - \mu\right)\right)^2\right] \\ &= \mathbb{E}\left[\left(\widehat{\mu}_{\Delta t,M} - \mu_{\Delta t}\right)^2\right] + \mathbb{E}\left[\left(\mu_{\Delta t} - \mu\right)^2\right] + 2\underbrace{\mathbb{E}\left[\left(\widehat{\mu}_{\Delta t,M} - \mu_{\Delta t}\right)\left(\mu_{\Delta t} - \mu\right)\right]}_{=0} \end{aligned}$$

$$= \mathrm{Var}\left(\widehat{\mu}_{\Delta t,M}\right) + \mathrm{Bias}\left(\widehat{\mu}_{\Delta t,M},\mu\right)^2$$

In this case, the variance is the Monte Carlo error and the $Bias^2$ is the Δt error. Now we have that:

$$\mathrm{Bias} = \mathcal{O}\left(\Delta t^{\alpha}\right) \tag{3.97}$$

and,

$$\mathrm{Var}\left(\widehat{\mu}_{\Delta t,M}\right) = \mathrm{Var}\left(\frac{1}{M}\sum_{i=1}^{M}\phi\left(Y_N^{(i)}\right)\right) = \frac{1}{M^2}\mathrm{Var}\left(\sum_{i=1}^{M}\phi\left(Y_N^{(i)}\right)\right) = \frac{1}{M}\mathrm{Var}\left(\phi(Y_N)\right). \tag{3.98}$$

Therefore we obtain:

$$\mathrm{MSE}\left(\widehat{\mu}_{\Delta t,M}\right) = \frac{1}{M}\mathrm{Var}\left(\phi(Y_N)\right) + \mathrm{Bias}\left(\widehat{\mu}_{\Delta t,M},\mu\right)^2 = \mathcal{O}\left(\frac{1}{M}\right) + \mathcal{O}\left(\Delta t^{2\alpha}\right). \tag{3.99}$$

□

In general, a very good result would be to have $\mathrm{MSE} = \mathcal{O}(\epsilon^2)$. Then by the above we would require $M = \mathcal{O}(\epsilon^{-2})$ and $\Delta t = \mathcal{O}(\epsilon^{1/\alpha})$.
We finally make a remark on the overall cost C of some algorithms in this case. Note that the dominant cost corresponds to the generation of paths. We have:

$$C\left(\widehat{\mu}_{\Delta t,M}\right) = \mathcal{O}\left(M\Delta t^{-1}\right) = \mathcal{O}\left(\epsilon^{-2-1/\alpha}\right). \tag{3.100}$$

Now, in the case of an Euler-Maruyama Scheme and for $\alpha = 1$,

$$C\left(\widehat{\mu}_{\Delta t,M}\right) = \mathcal{O}\left(\epsilon^{-3}\right). \tag{3.101}$$

And in the case of a Milstein Taylor Scheme and for $\alpha = 2$,

$$C\left(\widehat{\mu}_{\Delta t,M}\right) = \mathcal{O}\left(\epsilon^{-2.5}\right). \tag{3.102}$$

3.8 Multilevel Monte Carlo Simulation (MLMC Simulation)

In this section, we aim to reduce the overall cost C of the simulation. The idea is to introduce a sequence of discretizations $\Delta t_0 > \Delta t_1 > ... > \Delta t_L$ with $\Delta t_L = \mathcal{O}\left(\epsilon^{1/\alpha}\right)$ is the desired time step (e.g. $\Delta t_l = \Delta t_L 2^{L-l}$). Now:

$$\begin{aligned}\mathbb{E}[\phi(X_T)] &\approx \mathbb{E}\left[\phi\left(Y_{N_L}^L\right)\right] \\ &= \mathbb{E}\left[\phi\left(Y_{N_L}^L\right) - \phi\left(Y_{N_{L-1}}^{L-1}\right)\right] + \mathbb{E}\left[\phi\left(Y_{N_{L-1}}^{L-1}\right) - \phi\left(Y_{N_{L-2}}^{L-2}\right)\right] + ... + \mathbb{E}\left[\phi\left(Y_{N_0}^0\right)\right] \underbrace{-\mathbb{E}\left[\phi\left(Y_{N_{-1}}^{-1}\right)\right]}_{=0}.\end{aligned}$$

Therefore we obtain:

$$\mathbb{E}\left[\phi\left(Y_{N_L}^L\right)\right] = \sum_{l=0}^{L}\mathbb{E}\left[\phi\left(Y_{N_l}^l\right) - \phi\left(Y_{N_{l-1}}^{l-1}\right)\right], \tag{3.103}$$

and for independent $\left(Y_{N_l}^{l,(i,l)}, Y_{N_{l-1}}^{l-1,(i,l)}\right)$ we let,

$$\widehat{\mu}_{\text{MLMC}} = \sum_{l=0}^{L} \frac{1}{M_l} \sum_{i=1}^{M_l} \left(\phi\left(Y_{N_l}^{l,(i,l)}\right) - \phi\left(Y_{N_{l-1}}^{l-1,(i,l)}\right)\right). \tag{3.104}$$

Now, similarly to the previous section, we can perform a mean-square error analysis.

Proposition 3.8.1. *The Multilevel Mean-Squared Error is given by:*

$$\text{MSE}\left(\widehat{\mu}_{\text{MLMC}}\right) := \mathbb{E}\left[\left(\widehat{\mu}_{\text{MLMC}} - \mu\right)^2\right] = \sum_{l=0}^{L} \frac{V_l}{M_l} + \mathbb{E}\left(\phi(X_T) - \phi\left(Y_{N_L}^{L}\right)\right)^2, \tag{3.105}$$

where $V_l = \text{Var}\left(\phi\left(Y_{N_l}^{l}\right) - \phi\left(Y_{N_{l-1}}^{l-1}\right)\right)$.

Proof. We first observe that:

$$\begin{aligned}
\mathbb{E}\left(\widehat{\mu}_{\text{MLMC}}\right) &= \mathbb{E}\left[\sum_{l=0}^{L} \frac{1}{M_l} \sum_{i=1}^{M_l} \left(\phi\left(Y_{N_l}^{l,(i,l)}\right) - \phi\left(Y_{N_{l-1}}^{l-1,(i,l)}\right)\right)\right] \\
&= \sum_{l=0}^{L} \mathbb{E}\left[\phi\left(Y_{N_L}^{l}\right) - \phi\left(Y_{N_{L-1}}^{l-1}\right)\right] = \mathbb{E}\left[\phi\left(Y_{N_L}^{L}\right)\right].
\end{aligned}$$

Furthermore, we can find an expression for the variance:

$$\begin{aligned}
\text{Var}\left(\widehat{\mu}_{\text{MLMC}}\right) &= \text{Var}\left[\sum_{l=0}^{L} \frac{1}{M_l} \sum_{i=1}^{M_l} \left(\phi\left(Y_{N_l}^{l,(i,l)}\right) - \phi\left(Y_{N_{l-1}}^{l-1,(i,l)}\right)\right)\right] \\
&= \sum_{l=0}^{L} \frac{1}{M_l} \text{Var}\left[\phi\left(X_{N_L}^{l}\right) - \phi\left(Y_{N_{L-1}}^{l-1}\right)\right].
\end{aligned}$$

Hence for $\text{Bias}(\widehat{\mu}_{\text{MLMC}}) = \mathbb{E}\left[\phi(X_T) - \phi\left(Y_{N_L}^{L}\right)\right]$ and $V_l = \text{Var}\left(\phi\left(Y_{N_l}^{l}\right) - \phi\left(Y_{N_{l-1}}^{l-1}\right)\right)$ we finally obtain that:

$$\text{MSE}\left(\widehat{\mu}_{\text{MLMC}}\right) = \sum_{l=0}^{L} \frac{V_l}{M_l} + \underbrace{\mathbb{E}\left(\phi(X_T) - \phi\left(Y_{N_L}^{L}\right)\right)^2}_{\Delta t_L \text{ error} \sim \mathcal{O}(\Delta t^{2\alpha})}. \tag{3.106}$$

□

Now recall that $\text{Bias} = \mathcal{O}\left(\Delta t_L^{\alpha}\right)$ and we have:

$$\begin{aligned}
V_l = \text{Var}\left(\phi\left(Y_{N_l}^{l}\right) - \phi\left(Y_{N_{l-1}}^{l-1}\right)\right) &\leq \mathbb{E}\left[\left(\phi\left(Y_{N_l}^{l}\right) - \phi\left(Y_{N_{l-1}}^{l-1}\right)\right)^2\right], \qquad \phi\text{-Lipschitz} \\
&\leq L^2 \mathbb{E}\left[\left|Y_{N_l}^{l} - Y_{N_{l-1}}^{l-1}\right|^2\right] \\
&= L^2 \mathbb{E}\left[\left|\left(Y_{N_l}^{l} - X_T\right) + \left(X_T - Y_{N_{l-1}}^{l-1}\right)\right|^2\right] \\
&\leq 2L^2 \left(\mathbb{E}\left[\left|Y_{N_l}^{l} - X_T\right|^2\right] + \mathbb{E}\left[\left|X_T - Y_{N_{l-1}}^{l-1}\right|^2\right]\right) \\
&\leq \mathcal{O}\left(\Delta t_l^{2q}\right) + \mathcal{O}\left(\Delta t_{l-1}^{2q}\right) \\
&= \mathcal{O}\left(\Delta t_l^{\beta}\right)
\end{aligned}$$

where $\beta = 2q$ and $\Delta t_{l-1} = s\Delta t_l$ for $s > 1$. Now if we want as before $\text{MSE} = \mathcal{O}(\epsilon^2)$, assuming that $\Delta t_l = Cs^{-l}$ for $s > 1$ we impose:

$$\text{Bias}^2 = \mathcal{O}\left(\Delta t_L^{2\alpha}\right) = \mathcal{O}\left(\epsilon^2\right) \tag{3.107}$$

and,

$$s^{-L\alpha} = \mathcal{O}(\epsilon) \Longrightarrow L = \frac{1}{\alpha \log(s)} |\log(\epsilon)|. \tag{3.108}$$

Remark that a more optimal tuning would be:

$$\frac{V_l}{M_l} = \mathcal{O}\left(\frac{\epsilon^2}{L+1}\right) \Longrightarrow M_l \approx (L+1)\epsilon^{-2} s^{-l\beta} \tag{3.109}$$

and then, for a general γ:

$$\text{Cost}(\widehat{\mu}_{\text{MLMC}}) = \sum_{l=0}^{L} M_l \Delta t^{-\gamma} \approx \sum_{l=0}^{L} (L+1)\epsilon^{-2} s^{-l\beta + l\gamma} = (L+1)\epsilon^{-2} \sum_{l=0}^{L} s^{-l(\beta-\gamma)}. \tag{3.110}$$

$\boxed{\beta > \gamma}$ We have:

$$\text{Cost}(\widehat{\mu}_{\text{MLMC}}) = \mathcal{O}\left(L\epsilon^{-2}\right) = \mathcal{O}\left(|\log(\epsilon)|\epsilon^{-2}\right). \tag{3.111}$$

$\boxed{\beta = \gamma}$ We have:

$$\text{Cost}(\widehat{\mu}_{\text{MLMC}}) = \mathcal{O}\left(L^2\epsilon^{-2}\right) = \mathcal{O}\left(|\log(\epsilon)|^2\epsilon^{-2}\right). \tag{3.112}$$

$\boxed{\beta < \gamma}$ We have:

$$\text{Cost}(\widehat{\mu}_{\text{MLMC}}) = \mathcal{O}\left(L\epsilon^{-2} s^{-L(\beta-\gamma)}\right) = \mathcal{O}\left(L\epsilon^{-2}\epsilon^{-\frac{\gamma-\beta}{\alpha}}\right) = \mathcal{O}\left(|\log(\epsilon)|\epsilon^{-2-\frac{\gamma-\beta}{\alpha}}\right). \tag{3.113}$$

Remark that for the Euler-Maruyama Scheme, $\alpha = \beta = \gamma = 1$, and we obtain:

$$\text{Cost}(\widehat{\mu}_{\text{MLMC}}) = \mathcal{O}\left(|\log(\epsilon)|^2\epsilon^{-2}\right). \tag{3.114}$$

Remark 3.8.2. *The optimal tuning is:*

$$\{M_l^*\} = \arg\min \sum_{l=0}^{L} M_l \Delta t_l^{-\gamma}, \quad \textit{such that MSE} = \mathcal{O}(\epsilon^2). \tag{3.115}$$

And for this choice we set:

$$\text{Cost}(\widehat{\mu}_{MLMC}) = \begin{cases} \mathcal{O}\left(\epsilon^{-2}\right), & \beta > \gamma \\ \mathcal{O}\left(\epsilon^{-2}\log(\epsilon)\right), & \beta = \gamma \\ \mathcal{O}\left(\epsilon^{-2-\frac{\gamma-\beta}{\alpha}}\right), & \beta < \gamma \end{cases}. \tag{3.116}$$

Finally note that we always have $\alpha \geq \beta/2$ which leads to $\epsilon^{-2-\frac{\gamma-\beta}{\alpha}} \leq \epsilon^{-2-\frac{\gamma}{\alpha}}$.

3.8.1 Unbiased Multilevel Monte Carlo Simulation

Recall that in a Multilevel Monte Carlo,

$$\mathbb{E}[\phi(X_T)] \approx \mathbb{E}\left[\phi\left(Y^L_{N_L}\right)\right] = \sum_{l=0}^{L} \mathbb{E}\left[\phi\left(Y^l_{N_l}\right) - \phi\left(Y^{l-1}_{N_{l-1}}\right)\right]. \tag{3.117}$$

In the unbiased case, we instead consider probability mass functions $(p_0, p_1, ...)$ where $p_i > 0$ and $\sum_{i=0}^{\infty} p_i = 1$. Now, setting $\lambda \sim (p_0, p_1, ...)$ and W as a Brownian Motion we have the following unbiased MLMC estimator:

$$\begin{aligned}
\mathbb{E}[\phi(X_T)] = \sum_{l=0}^{\infty} p_l \left(\frac{\mathbb{E}\left[\phi\left(Y^l_{N_l}\right) - \phi\left(Y^{l-1}_{N_{l-1}}\right)\right]}{p_l} \right) &= \mathbb{E}_{\lambda\sim p}\left(\mathbb{E}_W\left(\frac{1}{p_\lambda}\left[\phi\left(Y^\lambda_{N_\lambda}\right) - \phi\left(Y^{\lambda-1}_{N_{\lambda-1}}\right)\right]\right)\right) \\
&= \mathbb{E}_{\lambda,W}\left(\frac{1}{p_\lambda}\left[\phi\left(Y^\lambda_{N_\lambda}\right) - \phi\left(Y^{\lambda-1}_{N_{\lambda-1}}\right)\right]\right) \\
&\approx \frac{1}{M}\sum_{i=1}^{M} \frac{1}{p_{\lambda^i}}\left[\phi\left(Y^{\lambda^i}_{N_{\lambda^i}}(W^i)\right) - \phi\left(Y^{\lambda^i-1}_{N_{\lambda^i-1}}(W^i)\right)\right] \equiv \widehat{\mu}_{\text{UMLMC}}.
\end{aligned}$$

However, the cost is now a random variable and one can verify that,

$$\text{Var}(\widehat{\mu}_{\text{UMLMC}}) = \frac{1}{M}\sum_l \frac{V_l}{p_l} \quad \text{and} \quad \mathbb{E}\left(\text{Cost}(\widehat{\mu}_{\text{UMLMC}})\right) = M\sum_l p_l \Delta t_l^{-\gamma}. \tag{3.118}$$

These two quantities behave as,

$$\text{Var}(\widehat{\mu}_{\text{UMLMC}}) = \frac{1}{M}\sum_l p_l \Delta t_l^{\beta} \quad \text{and} \quad \mathbb{E}\left(\text{Cost}(\widehat{\mu}_{\text{UMLMC}})\right) = M\sum_l p_l \Delta t_l^{-\gamma}. \tag{3.119}$$

Hence this method only works if $\beta > \gamma$ with $p_l \sim \Delta t_l^{\delta}$, for $\gamma < \delta < \beta$, otherwise there does not exist a probability mass function p_l such that it makes $\text{Var}(\widehat{\mu}_{\text{UMLMC}})$ and $\mathbb{E}\left(\text{Cost}(\widehat{\mu}_{\text{UMLMC}})\right)$ finite quantities.